



Trimble® Earthworks Dozer

For SITECH Only

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Contents

1	Dozer commissioning checklists	7
1.1	Checklist—Pre-installation	7
1.2	Checklist—Configuring a system	8
2	Configuration	10
2.1	Install the firmware onto EC520	10
2.1.1	Gather required firmware	10
2.1.2	Obtain the required licenses	11
2.1.3	Connect to the EC520	11
2.1.4	Power-up the EC520 controller	11
2.1.5	Connect to the EC520 controller via the Web Interface	12
2.2	Recommended TD5x0 firmware versions	12
2.3	Storage space requirement for firmware upgrades	13
2.4	Over-the-air (OTA) firmware updates	13
2.4.1	OTA firmware update troubleshooting	14
2.5	Automatic upgrade of devices	14
2.6	Configure software using the Web Interface	15
2.6.1	Set up the Wi-Fi network	15
2.6.2	Complete the Configure section	16
2.6.3	Check GNSS components are connected	17
2.6.4	Customize the MSS band settings - if needed	19
2.6.5	Start using the TD5x0 display	19
2.6.6	Upgrade or downgrade the EC520 firmware	20
2.6.7	Upgrade the MS9xx GNSS receiver with the minimum firmware	21
2.6.8	Upgrade the GS5xx sensor firmware	21
2.6.9	Update the VM510 valve module and CAN Gateway	21
2.7	Tasks	22
2.7.1	Transferring Files	22
2.7.2	Export a file or folder to Connected Community	25
2.7.3	Convert an existing design file	25
2.7.4	Configure an avoidance zone	27
2.7.5	Start working on a supported third party tablet	28
2.7.6	Reset software to default settings	28
2.7.7	TD5x0 Recovery Mode	28
2.7.8	Exit Recovery Mode menu	29
2.7.9	Configuring a serial radio	29
2.7.10	Configuring a generic Ethernet gateway	30
2.8	Set up licenses for this machine	30
2.8.1	Introduction to licenses	30
2.8.2	Apply required core licenses	31
2.8.3	Updating licenses with an internet-connected machine	31
2.8.4	Updating licenses manually	31
2.8.5	License errors	32
2.9	Online services	32

2.9.1	Configuring an online service	32
2.9.2	Additional options	33
2.9.3	WorksOS errors	34
2.9.4	Communication with Cat online services	34
2.10	Receiving projects and designs from WorksManager	36
2.10.1	Troubleshooting	36
2.10.2	Known issues	37
2.11	Syncing files with VisionLink Productivity	38
2.12	Using coordinate systems	38

3 Dozer measure-up best practices guide 40

3.1	Overview of dozer measure-up	40
3.1.1	Purpose of this guide	40
3.1.2	Prerequisites	40
3.1.3	Measure-up workflow	41
3.1.4	Equipment list	41
3.1.5	Machine setup	41
3.2	Check machine wear	41
3.2.1	Introduction	41
3.2.2	Position the machine	42
3.2.3	Check the cutting edge	42
3.3	Preparing for measure-up	42
3.3.1	Overview	42
3.3.2	Set up the machine for measurement	42
3.4	Mark measure-up points	43
3.4.1	Introduction	43
3.4.2	Best practices	44
3.4.3	Mark blade measurement points	44
3.4.4	Mark cab-mounted GNSS measurement points (AR, AL, QR, QL and T)	48
3.4.5	Mark blade-mounted mast measurement points (MX, FX, GX)	49
3.4.6	Mark track edge measurement points (TF and TB)	51
3.5	Position the total station instrument	52
3.5.1	Measure cab-mount GNSS receivers with Active Body IMU sensors	52
3.5.2	Measure cab-mount GNSS receivers with Inactive Body IMU sensors	53
3.5.3	Measure blade-mount GNSS receivers	53
3.5.4	Cab-mount GNSS receivers with Inactive Body IMU sensors	54
3.6	Measure-up	54
3.6.1	Overview	54
3.6.2	Measurement workflow	54
3.6.3	Measure the C-frame or push arm angle	55
3.6.4	Measure the blade pitch	56
3.6.5	Take digital level readings for machine mainfall	57
3.6.6	Entering the values into the Measure-up section in the Web Interface	57
3.7	Using a USB flash drive to transfer measure-up data in .csv format	57
3.7.1	Prerequisites	57
3.7.2	Exporting the machine measurements as a .csv file	58
3.7.3	If required, edit the .csv file	58

3.7.4	Import the .csv file into the Web Interface	59
4	Dozer measure-up and sensor calibration validation guide	60
4.1	Overview of dozer measure-up and sensor calibration validation	60
4.1.1	Purpose of this guide	60
4.1.2	Prerequisites	60
4.1.3	Equipment list	61
4.2	Check static blade guidance on level ground	61
4.2.1	Check - Active Body IMU machines	61
4.2.2	Check - Inactive Body IMU machines	62
4.2.3	Corrective action	62
4.3	Check static blade guidance on a slope	62
4.3.1	Prerequisite	62
4.3.2	Check	63
4.3.3	Corrective action	63
4.4	Check dynamic blade guidance on a slope	63
4.4.1	Complete the Sensor Correction Check	63
4.4.2	Check	63
4.4.3	Corrective action	64
4.5	Save and activate machine files	64
4.5.1	Save a machine file	64
4.5.2	Activate, or restore, a machine file	64
5	Dozer valve calibration setup guide – Inactive Body IMU machines	66
5.1	Overview of dozer valve calibration setup	66
5.2	Measure the blade arm pivot length – Inactive body IMU machines only	66
5.3	Set the blade position	67
5.4	Mount a sensor for valve calibration	68
6	Dozer Autos tuning guide	70
6.1	Overview of valve tuning	70
6.1.1	Purpose of this guide	70
6.1.2	Overview of tuning	70
6.1.3	Prerequisites	71
6.2	Tune the lift valves	71
6.2.1	Prepare for tuning	72
6.2.2	Set the unloaded raise and lower speeds	72
6.2.3	Perform a test pass	73
6.2.4	Find the performance transition speeds	73
6.2.5	Tune the valves	74
6.3	Tune the tilt valves	75
6.3.1	Overview	75
6.3.2	Set the tilt speed	75
6.4	Save Settings	75

7	Troubleshooting decision tree	76
7.1	Check physical	76
7.2	Check electrical	78
7.2.1	TD5x0 display does not turn on	78
7.2.2	Further tests	79
7.2.3	Other electrical issues	80
7.2.4	MT9xx machine target status indicators troubleshooting	80
7.2.5	GS5x0 sensor troubleshooting	81
7.3	Check configuration	82
7.3.1	Check compatibility of GNSS receivers	86
7.3.2	Replace GNSS receivers	88
7.3.3	License troubleshooting	88
7.4	Check accuracy	89
7.4.1	2D-Sensor troubleshooting	91
7.4.2	2D system troubleshooting	92
7.5	Check performance	93
7.6	System errors	93
7.6.1	GNSS receiver errors	93
7.6.2	WorksOS errors	95
7.6.3	License errors	97
7.6.4	Device errors	97
7.6.5	Design validation	99
7.6.6	TCC (Connected Community) errors	106
8	TD5x0 Android Recovery Mode	111
8.1	Recovery Mode	111
8.2	Reset a TD510 or TD520 Android password	111
8.3	Reset a TD540 Android password	111
8.4	Exit the No Command screen	112
8.5	Exit Recovery Mode menu	112
9	Supported third party tablet installation	113
9.1	Overview of third-party tablet installation	113
9.1.1	Purpose of this guide	113
9.1.2	Supported third-party tablets	113
9.1.3	Prerequisites	113
9.1.4	Equipment list	114
9.2	Third-party tablet installation	114
9.2.1	Download the app to the tablet	114
9.2.2	Locate and attach the tablet	114
9.2.3	Validate the install	115
10	Tools and other supplies	116
10.1	Dozer measure-up best practices guide	116
10.2	Dozer measure-up and sensor calibration validation guide	116
10.3	Supported third party tablet installation	116

11 Torque charts	117
11.1 Imperial	117
11.2 Metric	117
12 Notices	119

Dozer commissioning checklists

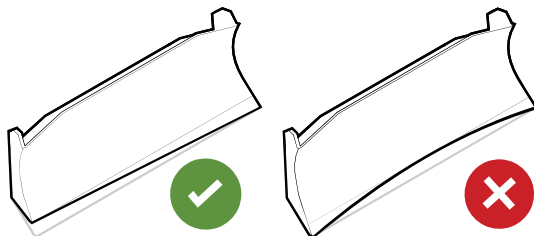
In this guide:

- ▶ Checklist—Pre-installation
- ▶ Checklist—Configuring a system

1.1 Checklist—Pre-installation

Use the following checks to identify problems with a machine before an installation. Problems that are identified by this procedure must be corrected before commissioning is begun. Poor machine condition can affect the performance of the automatic blade guidance.

- ☐ Operate the machine and observe the performance of the hydraulic system, particularly the responsiveness of the hydraulic system to control movements. Any hydraulic system issues will remain after installation, and may be worsened if an Autos system is installed.
- ☐ Check the cutting edge:
 - Even vertical wear over the entire length of the blade. 0 ± 5 mm horizontal deviation.



- ☐ Check the track tension: refer to the machine manufacturer's operator manual for guidance.
- ☐ Check the hydraulic rams and joints: place the blade on flat ground. Activate the rams to raise the blade and ensure there is limited play in the joints.
- ☐ On C-frame machines, check for wear on the ball joint.
- ☐ On EH machines, for machine ECM firmware:
 - Contact the machine manufacturer to ensure that the ECM firmware is up to date.

- ❑ On EH machines, perform a:
 - Joystick calibration
 - Solenoid or cracking current calibration

NOTE –Front idler height can also affect Autos performance. This requires dealer support to assess and adjust.

1.2 Checklist—Configuring a system

- ❑ Connect all system components:
 - EC520
 - Display
 - GNSS receivers
 - MT900 target (optional)
 - LR410 laser receiver (optional)
 - EM410 mast (optional)
 - Remote switch (optional)
 - Internet gateway (optional)
 - Radio (optional)
 - The GS5xx grade slope sensors appropriate to your system
- ❑ Turn on the ignition to start the display and to provide power to the EC520.
- ❑ Confirm that the display's date and time are correct. To change the date and time, open Settings > System > Date & Time > Time Zone and select the correct time zone. Complete the EC520 firmware installation as per the release notes.
- ❑ Tap the Web Interface icon and log in.
- ❑ If this is the first Web Interface log in:
 - set an admin password
 - configure Wi-Fi
 - complete the setup section of the Install Assistant

Otherwise, if this is a subsequent Web Interface login, the home page opens.
- ❑ Open Advanced > Licenses and apply a core bundle license. You can also apply any module licenses for the EC520.
- ❑ Open Configure > Install Assistant and complete each section. Use a TD5x0 or Wi-Fi BYOD display to perform valve calibrations.
- ❑ If using GNSS for 3D guidance, select Network > GNSS Correction Source and select a correction source.
- ❑ If using GNSS for 3D guidance, select Monitor > GNSS Details and check the GNSS receiver.
- ❑ Close the Web Interface, open the Operator App, log in and log out again. If the Operator App launcher is not installed, use MC Installer to install it.
- ❑ Open the Web Interface's Advanced > Licenses screen and apply the remaining module licenses to the display.

- ☐ If you have a WorksManager/Connected Community subscription:
 - Select Network > GNSS Connected Community; set up WorksManager/Connected Community.
 - Upload project files to WorksManager/Connected Community to sync to the machine.
 - Select Operation > Projects and set up the interval the EC520 synchronizes with WorksManager/Connected Community.
- ☐ Close the Web Interface again, and re-open the Operator App.
- ☐ If you are using a USB flash drive to transfer project files onto the system, insert and transfer from the USB flash drive to the machine.
- ☐ Confirm that all tiles on the Dashboard are not red. The Start button is then enabled, and tapping the button opens the work screen.
- ☐ Touch and hold on the text ribbon to configure the text items.
- ☐ Confirm that guidance is correct:
 - 2D – confirm system accuracy using the steps in the machine-specific *Measure-up and Sensor Calibration Validation Guide*.
 - 3D – confirm the Northing, Easting and Elevation position at the Focus tip.

Test the system gives accurate position while working on a design.

Configuration

In this chapter:

- ▶ Install the firmware onto EC520
- ▶ Recommended TD5x0 firmware versions
- ▶ Storage space requirement for firmware upgrades
- ▶ Over-the-air (OTA) firmware updates
- ▶ Automatic upgrade of devices
- ▶ Configure software using the Web Interface
- ▶ Tasks
- ▶ Set up licenses for this machine
- ▶ Online services
- ▶ Receiving projects and designs from WorksManager
- ▶ Syncing files with VisionLink Productivity
- ▶ Using coordinate systems

2.1 Install the firmware onto EC520

NOTE –The following procedures assume the use of a TD5x0 display as the upgrade device. If significantly different steps are required for a PC, the differences are shown as a Tip.

2.1.1 Gather required firmware

Download the latest .sg6 firmware from [Partners](#) using:

- USB flash drive – copy the firmware onto a USB flash drive. Insert the USB flash drive into the TD5x0 display's USB port.
- PC – copy the firmware onto the PC.
- TCC – copy the firmware to the machine-specific Trimble Connected Community EarthworksData folder. Synchronize the EC520 with TCC.

Apply the firmware in the Web Interface on Monitor > Onboard Devices. This process takes more than five minutes to complete.

NOTE – This firmware upgrade is for 2.24.x. Once the firmware is upgraded to 2.24.x, the system will limit how far you can downgrade the firmware in future.

NOTE –If using WorksManager to remotely apply firmware, we recommend that you copy the firmware into the machine-specific EarthworksData folder. Avoid using the Remote EC520 portal.

NOTE –When you use an SG6 file to update the software on your EC520, it replaces the files ww15mgh.ggf and xintcoordsys.xml. If you are using a custom geoid file named ww15mgh.ggf, the file will be replaced which may affect guidance.

2.1.2 Obtain the required licenses

Contact your SITECH dealer to purchase the licenses required for the installed system:

- Using a TD5x0 display—copy the licenses onto a USB flash drive. Insert the USB flash drive into the display's USB port and turn on the display.
- Using a PC—copy the licenses onto the PC.
- Using Cloud Licensing—In the Web Interface, turn on Auto Sync or use the Sync Now button on the Licenses page.

At a minimum, before you can install new EC520 system ECM firmware, *you must have*:

- A Core License installed.
- The Core license must provide a software maintenance date **ON** or **AFTER** March 2026. To determine the software maintenance date, refer to Software Maintenance Until item in the Web Interface > Advanced > Licenses screen.
- Completed the First Time Setup.

For information on installing licenses, refer to [Chapter 2.8, Set up licenses for this machine](#).

TIP – The firmware installation requires a Wi-Fi or physical connection to the ECM. If the ECM's Wi-Fi access point is not enabled:

- Use the TD5x0 display—connect the display to the harness in the usual way.
- Use a PC—use a standard Ethernet cable and an Ethernet Service Adapter (P/N150424-01) to connect the PC to the ECM.

2.1.3 Connect to the EC520

The firmware installation requires a physical connection to the EC520:

- Using the TD5x0 display – connect the display to the harness in the usual way.
- Using a PC – connect the PC to the EC520 using a standard Ethernet cable and the Ethernet Service Adapter (P/N 150424-01).

2.1.4 Power-up the EC520 controller

Connect the controller to a reliable power source.

2.1.5 Connect to the EC520 controller via the Web Interface

1. Connect the upgrade device (either a TD5x0 display or a PC) to the EC520 controller using the required cables.
2. Select the Web Interface icon on the TD5x0 display. The install screen appears.

TIP – Browse to the controller Web Interface IP address: 192.168.168.1. The install screen appears.

3. Select Choose File:
 - If using a PC – browse to the downloaded firmware file on the PC's file system, and select the file.
 - If using the TD5x0 display:
 - a. Select Documents. The Recent screen appears.
 - b. Select the menu icon in the top right corner of the screen.
 - c. Select Show SD card.
 - d. The USB flash memory appears as a folder in the left hand menu. Browse to the new firmware file and select the file.
4. Select Install OS. Firmware upload begins.
5. When the upload completes the firmware installation begins, and a progress screen appears.
6. Once the installation completes and the controller finishes rebooting, select the Web Interface icon. The Web Interface appears.

TIP – Once the installation completes and the controller finishes rebooting, browse to the controller Web Interface IP address: 192.168.168.1. The Web Interface appears.

2.2 Recommended TD5x0 firmware versions

The recommended firmware versions are:

- TD520 and TD510 - v4.31.2
- TD540:
 - Partitioned units - v002.000.015
 - Non-partitioned units - v002.001.015

The minimum firmware versions that are compatible with this release are:

- TD520 /TD510 - v4.31.1
- TD540:
 - Partitioned units- v002.000.015
 - Non-partitioned units - v002.001.015

Refer to the *TD5x0 release notes*, available within the TD5x0 OS Release Packages on [Partners](#), for more information.

2.3 Storage space requirement for firmware upgrades

Firmware upgrades require storage space to be available. The amount of storage space required is different for different firmware versions:

- v1.0.x - v2.14.x – at least 1.3 GB of available storage space
- v2.15.x onwards – at least 900 MB of available storage space

When not enough storage space is available, the system deletes old files until enough storage space becomes available. Files are deleted:

- One at a time
- In order by type: <fl2>, <s>, <partial>, <tmp>, <filepart>, <SG6> (at least one SG6 file is maintained), <zsnap>, <Archived_tag>, <Archived_pav>

2.4 Over-the-air (OTA) firmware updates

This section describes remote delivery of firmware updates.

WorksManager can push OTA ECM firmware file (.SG6) updates. This enables a technician to update a machine without being physically present.

When the OTA SG6 file is downloaded, the operator has 2 options:

- Install the OTA SG6 immediately
- Install the OTA SG6 at the next restart

The operator cannot repeatedly postpone the install. During installation, the Operator App and Web Interface are both locked out and show update progress. The SG6 is stored as a temporary file and is removed after installation. Due to the size of the SG6, it may take some time to download. If the download is interrupted, it resumes downloading from the point of interruption.

A technician can select an OTA SG6 file and apply it without waiting for the operator to acknowledge it. When the system restarts after installation, no further auto-install is required because the OTA SG6 is already applied.

When the ECM firmware is updated, individual devices on the system may no longer meet the minimum firmware required. Depending on the device, the system may update the device's firmware so the system can continue operating. See [Automatic upgrade of devices](#).

If the technician installs another SG6 in the presence of the OTA SG6, the OTA SG6 will be applied after reboot because the system detects the downloaded OTA SG6 at start-up. The technician should unassign the OTA SG6 in WorksManager if they want to activate another SG6.

When the update is complete, the system deletes the downloaded SG6 file to free up storage space.

You can still perform system updates by downloading an SG6 file to the system through the Web Interface.

2.4.1 OTA firmware update troubleshooting

If the ECM does not have enough free space, an error message displays. The system requires enough free space for both the SG6 and the files extracted from it. The system needs three to four times the SG6 size to download, uncompress and install the firmware file.

There is a Firmware Upgrades diagnostics section included in the diag.txt with the following information:

- Last downloaded firmware name
- Last downloaded firmware ID
- Time of last OTA server check
- Time of last firmware download
- Time of last firmware upgrade
- Latest OTA upgrade status
- Time of last successful upgrade

If your system is missing an up-to-date maintenance license when a new SG6 is assigned, a warning message displays and the install does not occur.

2.5 Automatic upgrade of devices

When some connected devices do not meet the minimum firmware requirements of the system, the device's firmware is automatically updated so the system can continue operating. This is generally triggered by:

- Minimum version requirements changing in newer system software
- A newly connected device not meeting that system's minimum firmware requirements

The Operator App and the Web Interface will show firmware update progress and provide the estimated time remaining. During this time, the system's Maintenance Lock is enabled. The technician can choose to override the automatic updates and cancel them. This gives the lock back to the technician.

The operator cannot cancel the updates because they are required to get a Start button with their current configuration and Machine Setup.

The device types that are updated when their firmware does not meet the minimum requirements are:

- SNR radio
- GS51x sensor
- GS520 sensor

- MS9x2 receiver
- MS9x5 receiver
- VM510 valve module

Devices that are not automatically updated and that do not meet the minimum firmware requirement must be updated separately.

2.6 Configure software using the Web Interface

2.6.1 Set up the Wi-Fi network

The EC520-W works wirelessly with any PC, laptop, tablet, or smart phone. Do the following:

1. Set up Wi-Fi:
 - a. Select the channel the office uses to communicate with this machine.
 - b. Enter your Service Set Identifier (SSID).
 - c. Enter a password in Password and Confirm Password fields.
 - d. Click Save.
2. If you want to confirm Wi-Fi is connected, open the Settings tab on other tablets or laptops and confirm they are connected to the EC520 network.
3. If required, open the Web Interface on your tablet or laptop and add permission to run the Operator App on your tablet.
 - a. Select Configure > Install Assistant > User Permissions.
 - b. Select Add to add an operator.
 - c. Enter the required details and click Save to save the user.
 - d. Tick the checkbox next to the operator to edit or delete the entry.
 - e. Toggle off the "Require Operator Password" option to disable operator password while operator log in. This option is toggled on by default.

TIP – For best performance when using Autos-related features, avoid running the Web Interface on a laptop connected to the system via Wi-Fi. Either run the Web Interface on the TD5x0 display or connect the laptop to the system with an Ethernet cable.

Some of the problems that you might see include:

Problem	Description	Resolution
With TD5x0 attached, the Wi-Fi Network page is not complete.	The Wi-Fi network on the EC520 is off, or the EC520 is not enabled for Wi-Fi.	With a laptop connected via Ethernet cable to the EC520 (or via the TD5x0), set up the machine in the Web Interface, then configure the Wi-Fi network as described above.
Wireless tablet cannot communicate with EC520.	The first time you try to communicate with a new EC520 you must be connected either by configuring the Wi-Fi or by using an Ethernet cable. Tablets do not have an Ethernet connection.	Configure your wireless tablet to connect to the EC520 Wi-Fi.

NOTE –The EC520-W can communicate with any Wi-Fi enabled device, but on the machine your TD5x0 display will, by default, use the Ethernet segment of the EC520 long platform harness to connect to the EC520. If the TD5x0 and the EC520 are not connected by an Ethernet cable, then the TD5x0 can be configured to communicate with the EC520 by Wi-Fi. This method of connection is not recommended for Autos-related functions.

2.6.2 Complete the Configure section

In the Web Interface complete all of the Configure section.

1. If using a computer (laptop connected by Ethernet or Wi-Fi), start the laptop.
 - a. Open Chrome (web browser).
 - b. Enter 192.168.168.1.
 - c. Skip to step 3.
2. If using a TD5x0 display, as soon as you supply power it starts and the Welcome screen appears.
3. Tap **Got It** to continue and complete the Android tour screens.
4. Tap the Web Interface icon (shovel and wrench).
5. Log in to the Web Interface using the default username (admin) and create a password.
6. Wait for the initial install wizard to open at the Setup screen and complete the Setup tasks.

NOTE –If steering control is enabled for dozers, enable the toggle on the Pass Shift page. This allows operators to use Pass Shift via either an installed horizontal offset remote switch or the Pass Shift text item in the Operator App.

NOTE –Using punctuation in the machine name is unsupported and may cause unexpected behavior. Avoid using punctuation in machine names.

7. When finished, the Install Assistant page opens. Acquire the maintenance lock and complete:
 - Install Assistant:
 - Sensor Correction Check - do this check first, if available
 - Measure-up
 - External Lightbars - optional
 - Wi-Fi Network

Things to note:

- Enable the external lightbars. Within a few seconds the lightbars, if connected, flash in sequence. For information on the lightbar start-up sequence, refer to the *External Lightbars* guide in the overflow menu of the Operator App or on [Partners](#).
- Complete the Remote Switches wizard. On dual CI520 interface module configurations, if the CI520s are disconnected from the harness after completing the Remote Switches wizard, redo the Remote Switches wizard to confirm that the remote switches are working as initially configured.

2.6.3 Check GNSS components are connected

After installing the hardware, check the following key components are connected, if required:

- Radio or internet gateway (receiving GNSS corrections)
- GNSS receivers (receiving position from satellites)

Do the following:

1. Open the Web Interface.
2. Select Network > GNSS Correction Source. The GNSS Correction Source screen shows the status of the radios.
3. If the screen shows "Not Connected" status, resolve the problem.
4. Activate and edit the desired GNSS Correction Source, and configure the settings for your site.
5. If the settings will not change, resolve the problem.
6. Select Monitor > Connected Devices, review the GNSS device information, and look for problems. For example, if RTK is not present then the system is not receiving corrections.
7. Resolve any problems.

At this stage your system will have an accurate position and be receiving GNSS corrections. Some of the problems that you might see include:

Problem	Description	Resolution
The GNSS Correction Source page shows the radio status is "Not Connected".	The grade control system uses the radio or internet gateway device to send GNSS corrections to the EC520. The Correction Source page in the Web Interface shows whether the radio is connected.	• Check the cables between the radio and EC520 are connected. • Check the radio firmware is up to date. Download the most recent firmware from your usual source.
The number of the channel on the radio will not change.	Once you have connected the radio, the one thing left to do is enter the channel that your site uses. If you cannot change the channel this indicates a problem.	Check the radio firmware is up to date.
The GNSS status is not RTK.	Once you have connected the GNSS receiver or receivers, and the site's positioning system is working, the GNSS receiver status will show as RTK. If it does not, this indicates a problem.	Check the positioning system: • Base station • Receiver firmware • Receiver hardware • Cabling

Problem	Description	Resolution
My laptop was connected to an EC520, but I moved and it isn't.	The laptop was connected to an EC520 via Wi-Fi, either directly or through docking station, but moving to another machine or undocking the laptop means the laptop disconnects.	Before taking the laptop to another machine or undocking, try using the keyboard commands Ctrl + Alt + Delete to secure your laptop. Or, reboot the laptop when at the new machine.
In the office my laptop was connected to an EC520, now it isn't.	You may have your laptop set up to connect automatically to the nearest Wi-Fi network. Bringing another EC520 near to your laptop may override the current connection.	In the task menu check the name of the network your laptop connected to. Also found at the top right hand corner of the Web Interface.
Earthworks on supported third party tablet will not appear.	Earthworks is installed but the way you logged in does not have rights to open Earthworks.	In the Web Interface, set -up permissions to allow you to use Earthworks.

2.6.4 Customize the MSS band settings - if needed

NOTE –These instructions are for machines using MS9xx receivers with firmware prior to v5.41. If using current firmware, leave the Use Custom toggle off as the latest receiver firmware has the correct frequencies and baud rates built in.

To customize the MSS band settings for receivers using old firmware:

1. Open Operation > GNSS Management
2. Enable the Custom RTX toggle
3. Update the values as required, frequencies listed in Trimble Bulletin

All settings on this page are sent to the MSS receiver or receivers on every system start up.

2.6.5 Start using the TD5x0 display

To set up Earthworks on the TD5x0 display:

1. Configure the system completely in the Web Interface, ensuring an operator is configured in the Web Interface under Configure > Install Assistant > User

Permissions.

2. Release the maintenance lock.
3. Supply power to the display.
4. Decide how you want to use the display, landscape (horizontal) or portrait (vertical). Rotate the display to this position. Tap the orientation button on the left side of the screen to adjust the screen orientation.
5. Select the language you want and then tap the arrow in the upper right corner.
6. Select your time zone and then tap the arrow in the upper right corner.
7. Set the date and time by scrolling each item and then tap the arrow in the upper right corner.
8. The Welcome screen appears. To change wallpapers, widgets, and settings, touch and hold the screen background.
9. Tap Got It to clear the Welcome screen.
10. If the display is running an older or newer version of the grade control system to the EC520, then upgrade/downgrade to the matching version.
 - a. On the Upgrade Required screen tap Download.
 - b. The Downloading percentage screen appears.
 - c. The Machine Control Plugin screen appears, tap Install.
 - d. On the App Installed confirmation screen, tap Done.
 - e. If you are prompted to install an update to the existing application, tap Install.
 - f. Tap the Earthworks icon.
 - g. When asked to allow the Machine Control Plugin to access files on the display, tap Allow.
 - h. Repeat steps a - c.
 - i. The **Allow the machine control plugin to access photos, media and files on your device** request appears. Tap Allow.
 - j. When you see the Viewing Full Screen panel on display screen, tap Got it.
 - k. Enter the name and password set in the Web Interface's Configure > Install Assistant > User Permissions and Login.
 - l. The Synchronizing... screen appears.
11. The Dashboard opens.

Earthworks now shows the current machine data. Finish configuring or installing to clear any error states.

2.6.6 Upgrade or downgrade the EC520 firmware

To upgrade or downgrade the EC520 system ECM firmware:

1. Open the Web Interface and browse to Monitor > Onboard Devices.
2. Select + Add Firmware.

3. Browse to the new firmware file, select it, and then select Open. The firmware starts uploading to the system ECM.
4. Wait for the upload to complete, then select the down arrow at the end of the system ECM row.
5. Select the new firmware and then select Apply Firmware. Once the installation completes the system ECM reboots.

NOTE –When the EC520 firmware is upgraded or downgraded, the machine settings files (*.machine) automatically save, preserving previously configured settings even when flashing issues occur. To access the automatically saved files, in the Web Interface, open Advanced > Machine Settings Files.

NOTE –When the EC520 firmware is downgraded, you must apply a machine settings file that is from the same downgraded version or earlier, or recommission the machine.

NOTE –If you want to downgrade from v2.3 or later to v2.0, you must first downgrade to either v2.2 or v2.1 and then downgrade to v2.0.

2.6.7 Upgrade the MS9xx GNSS receiver with the minimum firmware

To use 2.24.x and later of the grade control system, the following GNSS receivers must be upgraded with the minimum firmware:

- MS9x6 - firmware v5.56 or later
- MS9x5 – firmware v5.45 or later
- MS9x2 – firmware v5.31 or later

The latest required firmware is included in the software. After installing the software, in the Web Interface, open the Monitor > Onboard Devices to confirm the minimum firmware is installed.

2.6.8 Upgrade the GS5xx sensor firmware

When upgrading the firmware on a GS5xx sensor, avoid interrupting power to the sensor. Interrupting power may render the system incapable of identifying the sensor.

If power is interrupted, restart the system. If this does not make the sensor identifiable, contact your support channel.

2.6.9 Update the VM510 valve module and CAN Gateway

When upgrading motor graders, dozers or compact loaders from previous VM510 versions, the firmware must be updated to match the requirements below. The firmware is bundled with the software. To update:

1. Open the Web Interface's Monitor > Onboard Devices screen.
2. Select the available firmware for
 - a. Valve module - v1.02.10 or later
 - b. CAN Gateway - v1.04.00 or later
3. Tap Apply Firmware.

2.7 Tasks

This section describes some common tasks the installer may need to do.

2.7.1 Transferring Files

File Transfer screen

The File Transfer screen is available in the System Settings menu on the operator's interface. You can:

- Synchronize the files on the EC520 controller (the machine) with your organization on Connected Community (TCC).
- Import files to the machine from the display's hard drive or other external storage device.
- Export files from the machine to the display's hard drive or other external storage device.
- Custom file import files to the machine from the display's hard drive or other external storage device.

File transfer from an external storage device to a TD5x0

This section provides useful information about transferring files from an external storage device, such as USB flash drive, to a TD5x0 display.

For detailed information about synchronizing, importing and exporting files, refer to the operator's guide *Transferring Files*.

Requirements

You must have:

- a TD5x0 display
- an external storage device, such as a FAT32 format USB flash drive with a maximum capacity of 32GB

Files are overwritten with newest versions

When transferring files from an external storage device to the machine, all files on the machine that have the same name as files on the external storage device will be overwritten with the external storage device files. This includes the user preferences file `userdata.pref.xml`, which saves the settings; for example, the text items that are configured.

To keep the user preferences saved on a machine, either:

- Perform a File Transfer of the EarthworksData to the external storage device first, then perform a File Transfer from the external storage device to the machine

OR

- Untick the EarthworksData box when transferring files from the external storage device to the machine

NOTE –The userdata.pref.xml file is stored at ProjectLibrary > EarthworksData > [Machine Name Folder]

Folder structure

When using Import Files TO Machine, the required folder structure on the external storage device is:

- ProjectLibrary/Projects/<SiteName>/OfficeData/Designs

where:

Folder Name	Description
ProjectLibrary	Fixed root-directory folder name.
Projects	Fixed folder name.
SiteName	Name of the job site or project defined by user in the office software.
OfficeData	Fixed folder name. Includes: • coordinate systems (.cal) • site control points (.cpz) • site line work and site points (.smz) • avoidance zones (.avoid.svl) • feature codes (.fxl) • ground surface (.ttm) • current elevation (.cat)
Designs	Fixed folder name. Includes: • design files (.xml, .dsz and .vcl) • elevation design (.cat)

NOTE –Custom File Import TO Machine does not require the above folder structure.

File name rules

The system takes files that are in the OfficeData folder and uses them across the whole project. This includes Depth and Slope and Design guidance modes. The system uses only one file from each OfficeData category— .cal, .cpz, .smz, .avoid.svl, and .ttm. Use these file rules:

- .cal, .cpz and .smz - the system uses the highest version
- .avoid.svl - only one file within OfficeData is supported. Multiple files cause an error message to display on the operator interface when the project is loaded
- .ttm - the operator selects a file within the mapping dialog

File versions

File versions are set by the file transfer method used to transfer files to the OfficeData folder:

- Trimble Business Center - the system automatically adds a version number to the file name, using this format: *'file name'.Vxx.'file extension'*. Example: *NorthArea.V01.cal*
- Other method - the user decides whether to use this naming convention or not.

NOTE –The "V" is upper case.

In the OfficeData folder

In the Projects > OfficeData folder, the system reads both the file extension (.cal, .cpz, .smz) and the number (.Vxx) and then uses the following rules:

Description	Example
The system uses the largest .Vxx number associated with each file extension.	<i>NorthArea.V01.cal, NorthArea.V02.cal, SouthArea.V03.cal</i> The system uses <i>SouthArea.V03.cal</i>
If the .Vxx number AND the file extension are the same for two different files, the system displays an error.	<i>SouthArea.V01.cal, NorthArea.V01.cal</i> ".V01.cal" is the same for both files. The system displays an error and uses neither file until the user removes one file.
The system uses a file with a number in .Vxx format before a file without one.	<i>NorthArea.cal, NorthArea5.cal, NorthArea.V01.cal</i> The system uses <i>NorthArea.V01.cal</i>
If the file does not include .Vxx in the name, the system does not allow two files to have the same file extension.	<i>NorthArea.cal, SouthArea.cal</i> ".cal" is the same for both files. The system displays an error and uses neither file until the user removes one file.

In the OfficeData > Designs folder

In the Projects > OfficeData > Designs folder, the system reads both the file extension (.dsz) and the numbering system (.Vxx) and then uses the following rule:

Description	Example
All .dsz files appear on the display and the operator chooses which file to use.	<i>WestDesign.V01.vcl, EastDesign.V01.vcl, WestDesign.V02.dsz, EastDesign.V02.dsz</i>

2.7.2 Export a file or folder to Connected Community

There are two ways to export the Earthworks machine file or folders to Connected Community (TCC).

NOTE –This information applies only to users connected to Connected Community.

Export from Earthworks

1. Capture a backup of the complete system file structure.
2. Open Earthworks and tap Settings > File Transfer > Export from Display.

Export from the Web Interface

Complete one of the following:

- In the Web Interface, select Advanced > Machine Setting Files. Select the machine setting file you want to export, select the expanded menu button (bottom right), and select Download. A machine file has a .machine suffix.
- Capture a backup of the complete file structure along with current settings. In the Web Interface, select Operation > File Management and select Backup All. Upon completion, the backup file will be saved to the Download folder on the device that the Web Interface was being run on when the backup process was started, e.g., if you access the Web Interface via the TD5x0 and initiate the backup from there, on completion the backup file is saved to the Download folder on the TD5x0 and can be retrieved from there.

2.7.3 Convert an existing design file

The existing GCS900 or AccuGrade system uses a different type of design file to Earthworks. Use any of the following methods to set up the machine to use the correct type of design file.

Method 1: Use supported Trimble Business Center objects

Trimble Business Center supports the following Earthworks objects:

- Surface-like objects, including:
 - Surface
 - Corridor Surface
 - Depth Surface
 - Layer-based Surface
 - Over-excavation Surface
 - Planar Surface
 - Subgrade Surface
 - Topsoil Surface
 - Uncompacted PCS Surface
 - Trench Surface
- CAD linework objects, including:
 - Linestring
 - CAD 3D PolyLine
 - CAD PolyLine
 - CAD Arc
 - CAD Circle
 - CAD Line
 - CAD MultiLine Text
 - CAD Text
 - Horizontal Alignment
 - Breakline
 - Polygon
 - Boundary
 - Point and CAD Point - Marked on the background map, but not loaded as a guidable point

Method 2: Use Trimble Business Center

1. Open the existing Business Center Project file (.vce), which contains the existing design(s), using the latest version of Business Center.
2. If the original .vce project is not available:
 - a. Best practice is to import the raw design file.
 - b. Alternative method is to import the existing GCS900/AccuGrade design. Note that this may cause loss of some design detail such as alignment names.
3. Use Field Data > Publish to Library in Trimble Business Center to export the design for Earthworks vx.x. This will include the .dsz file, along with supporting coordinate system .cal file with GeoData folder, if required.

4. The files will sync with the EC520 at the time interval you set in "Transfer the files to the machine via Connected Community or USB/Display", as per the File Transfer process.

When you have completed the steps you will have a .dsz design file ready to use on the machine.

Method 3: Use File Flipper

File flipper converts GCS900/AccuGrade design files to Earthworks file types so design files can be re-used.

1. Transfer the design files from the CB4x0 onto an external storage device.
2. Use Trimble App Central or [Partners](#) to install File Flipper onto the TD5x0 or third party supported tablet.
3. Insert the USB flash drive into the TD5x0 and use File Flipper to convert the legacy design file to the .dsz file type.
4. Store the design in the Projects folder in the office software.

NOTE –Earthworks uses a .cal Site Calibration file that is stored in the OfficeData folder.

Method 4: Custom File Import TO Machine

Custom File Import TO Machine enables:

- File selection from an external storage device with any file folder structure
- Automatic conversion of .svd and .svl files to .dsz

When importing .svd and .svl files, both files must:

- have the same name
- be stored in the same folder

The Custom File Import TO Machine process then converts the .svd and .svl and creates one .dsz, which is then stored on the machine.

2.7.4 Configure an avoidance zone

Follow these rules to configure avoidance zones:

- Avoidance zones must be closed polygons.
- The maximum number of edges of all polygons in a file is 800.
- Export .avoid.svl files from Trimble Business Center using Machine Avoidance Zone exporter.
- Save .avoid.svl files to the OfficeData folder.
- Some measurements for machine boundaries in relation to avoidance zones are estimated. Confirm or update these measurements in Configure > Install Assistant > Additional Machine Dimensions.

NOTE –When an avoidance zone file has an error or warning, the operator can use the associated project and design files but avoidance zone guidance is not provided.

2.7.5 Start working on a supported third party tablet

Read the guide describing installing a supported third party tablet onto a generic excavator system, *Supported third party tablet install guide*.

2.7.6 Reset software to default settings

In certain circumstances you may need to complete a full factory reset of the EC520. Best practice is to Reset to Default when changing machine types or when downgrading the EC520 software. When permanently removing a Earthworks system, Reset to Default effectively clears all stored data.

Before starting, make a copy of all of the files in the file tab, or export a full copy of the machine files from the Earthworks app.

1. Open the Web Interface and select Advanced > Reset to Default.
2. Read the Reset to Default warning message, ensure your files are backed up and select Reset.
3. Type 'reset' into the Confirm Reset Of All Data, User Accounts, And Machine Measurements box and select Confirm Reset.

NOTE –Resetting to factory defaults does not affect licenses.

4. After resetting to factory defaults, the Web Interface opens:
 - a. Enter a new password
 - b. Accept the End User License Agreement
 - c. Configure the machine again

2.7.7 TD5x0 Recovery Mode

Recovery Mode

Recovery Mode is an option on the power down menu. When opened, Recovery Mode provides an option to clear a TD5x0 hard drive. This option is useful for resetting a TD5x0 password.

Reset a TD5x0 password

To reset the password on a TD5x0

1. Turn on the display. Press and hold the power button to open an options menu
2. Tap Recovery and then OK on the Recovery Mode box
3. Wait for the No Command, or dead Android, image to display
4. Press simultaneously the power and volume up button a few times until the Android Recovery menu opens

5. Follow the on-screen instructions to select wipe data/factory reset
6. Wait for the display to reboot and enter a new password

NOTE –Wipe data/factory reset erases all files, such as design files, from the display's hard drive. All installed apps remain.

Reset a TD540 Android password

To reset the Android password on a TD540:

1. Turn off the display. Press and hold the power and volume down buttons at the same time until the display turns on.
2. Wait while the device image displays.
3. Press the volume up button for five seconds and release. This makes the menu open.
4. Follow the on-screen instructions to select wipe data/factory reset.
Note - Tapping the screen also scrolls through the menu items.
5. Wait for the display to reboot and enter a new password.

NOTE –Wipe data/factory reset erases all files, such as design files, from the display's hard drive. All installed apps remain.

Exit the No Command screen

To exit the No Command, or dead Android, screen (before the Recovery Mode menu screen), press the display's power button or leave the display on for two minutes and it will automatically reboot.

2.7.8 Exit Recovery Mode menu

To exit the Recovery Mode menu, follow the on-screen instructions to select reboot system now.

2.7.9 Configuring a serial radio

The standard system radios are SNRxxx models. However, the system also supports generic serial radios for communication. If using a generic serial radio, configure it as follows:

Prepare the Generic (Serial) radio (required):

- Use the radio's own software to configure the radio separately. The Earthworks system cannot change a generic radio's settings.
- Set the baud rate to 38400 bits/second.
- Set the serial connection to 8 data bits, no parity, and 1 stop bit.

Enable the Generic (Serial) radio option in the system using the Web Interface (optional):

- If the system detects data on the auxiliary serial port (for example, corrections), it automatically enables the Generic (Serial) radio.

If desired, you can manually enable the Generic (Serial) radio option. From the Web Interface's Configure > Install Assistant screen, proceed through the Setup section and enable the Generic (Serial) toggle.

Select the Generic (Serial) radio as the active correction source in Earthworks using either the Operator App or the Web Interface (required):

- In the Operator App, select the Generic (Serial) radio as the active Correction Source from the Machine Setup tile.
- In the Web Interface, activate the Generic (Serial) radio on the Network > GNSS Correction Source screen.

2.7.10 Configuring a generic Ethernet gateway

The system supports generic Ethernet gateways. If using a generic Ethernet gateway, configure it as follows:

- Connect the generic gateway device to the system harness using the Internet ETH2 connector.
- Use the gateway device's own software to configure the device separately. The Earthworks system cannot change a generic gateway's settings.

Enable the generic gateway device in the system using the Web Interface:

1. From the Web Interface's Configure > Install Assistant screen, proceed through the Setup section to the Network devices screen.
2. Set the *Choose a device configuration* drop-down to Generic Gateway (Ethernet).

The following conditions apply:

- If the system is configured for an internet-based GNSS correction source, the generic Ethernet gateway is a required device.
- If the system does not require the internet for guidance, the device is listed as optional.
- If the system does not detect any Ethernet traffic, it lists the gateway device as missing.

2.8 Set up licenses for this machine

2.8.1 Introduction to licenses

A Software Maintenance License must be installed to use the system. You can manage licenses on the Web Interface's Advanced > Licenses screen. There are two types of Software Maintenance Licenses:

- Perpetual – an open-ended license. Stored on the EC520 controller or the TD5x0 display, and not transferable to another device.
- Subscription – a license that expires after a fixed term, if not renewed. Stored on the TD5x0 display or the GNSS receiver and not transferable to another device.

You can install licenses manually on your system, or sync them with the cloud.

The system provides a status icon beside each license:

- Red cross: You cannot operate the system with the current licenses. A license that is required to access the work screen is missing or expired.
- Orange exclamation mark: You can operate the system, but functionality may be limited or due to expire. Options include:
 - The license is currently usable, but is within its expiration warning period.
 - For the current configuration, a module license that is expected, and which adds functionality, is missing or expired. You can still access the work screen.
- Green tick: The license is installed correctly.

2.8.2 Apply required core licenses

The following core licenses must be applied before starting the following valve calibrations:

- Blade rotation calibration: Core License - Dozer
- Valve calibration: Core License - Dozer

2.8.3 Updating licenses with an internet-connected machine

NOTE –Your TD5x0 display must have firmware V4.30.0 or later.

If you have an internet connection, there are two ways to sync the licenses for your system with the cloud. Both options are on the Web Interface's Advanced > Licenses screen:

- If you enable the Auto Sync toggle, the system automatically synchronizes the licenses with the cloud every minute.
- If you select the Sync Now button, the system connects to the cloud and checks for new licenses once.

If the licenses installed on a device are from the cloud, a cloud icon shows beside the device . Once a device has cloud licenses loaded and this icon is present, it can no longer load or use previously-applied option keys. The Licenses screen shows the license synchronization status at the top of the screen.

Check that your licenses appear correctly.

NOTE –When you apply a subscription license to an MS9xx receiver, you cannot downgrade the firmware below version 5.54.

2.8.4 Updating licenses manually

When you manually install a license file:

- The system installs the license on the device covered by the serial number in the license file name.
- The file name is case sensitive. Ensure the extension is lowercase.

To install the license:

1. Make the license file accessible to the computer or device you are using to enter data: via a USB flash drive, on the device or on your network.
2. Open the Web Interface and select Advanced > Licenses.
3. Select Add License File and browse to the license file.
4. Select Add.
5. In the Advanced > Licenses screen, check that your licenses appear correctly.

2.8.5 License errors

For information on license error codes, see [License troubleshooting](#).

2.9 Online services

Depending on your system's configuration, you can connect to an online service:

- Trimble Construction Cloud
- VisionLink

An online service may provide benefits like automatic updates of projects/designs or automated system firmware updates.

2.9.1 Configuring an online service

To configure the connection to an online service, use the Web Interface's Network > Cloud Services screen.

Trimble Construction Cloud

Trimble Construction Cloud supports the following features:

Field	Description
Service Connection	Reports the status of the latest sync with the online service: • Green ticket: The most recent sync was successful • Grey icon: The system has not synced with the online service yet • Orange warning: The most recent sync failed Select the  icon to manually resync with the service.
WorksManager Reporting	Enables reporting position and status data to WorksManager.
WorksManager Firmware Over The Air	Enables WorksManager sending assigned firmware updates to the grade control system.

Field	Description
WorksOS Surface Download	Enables the operator to download a surface file (.ttm) from WorksOS: • In the Operator App's Cut/Fill Mapping screen, the operator can set the Ground Surface drop-down to WorksOS Download. • The machine's EC520 must be a registered device in your WorksOS account. • Your WorksOS account must have an active project for the machine's location.
Report Files (.tag)	Enables sending .tag files to WorksOS and VisionLink. If you disable this toggle, the .tag files are saved locally.
Remote Access	Enables remote access to the Web Interface.

VisionLink

VisionLink supports the following features:

Field	Description
Service Connection	Reports the status of the latest sync with the online service: • Green ticket: The most recent sync was successful • Grey icon: The system has not synced with the online service yet • Orange warning: The most recent sync failed
Remote Access	Enables remote access to the Web Interface.
Report Files (.tag)	Enables sending .tag files to VisionLink. If you disable this toggle, the .tag files are saved locally.

2.9.2 Additional options

The Additional fields enable the system to send anonymous reports about your machine to maintain, support and improve the software.

Field	Description
Crash Reports	Crash Reports contain crash and bug data about your software.
Usage Statistics	Usage Statistics send anonymous data about how operators use the app. This data helps to improve future releases.

2.9.3 WorksOS errors

See [WorksOS errors](#).

2.9.4 Communication with Cat online services

If the grade control system is connected to an A6N2 network manager (with the latest firmware: P/N 6467493-00.fl2) and a PL radio, the system will check that a valid Cat subscription exists. If the subscription is valid, the system can communicate information with VisionLink. This can extend some functionality.

If the system detects a base level subscription, the 4 core pieces of data that the system reports to the online service are:

- System is operating
- Power Down Delay
- License info
- PL radio signal strength

If the system detects a Cat Performance + Grade Connectivity subscription or a Performance Pro subscription, it also enables:

- Remote Access
- Cell GNSS corrections (VRS), if that is the configured correction source

If the system detects a Cat Performance Pro subscription, it also enables:

- Project Data:
 - Designs
 - Site Calibrations
 - Geo Data
- Tag files
- System snapshot
- Status Message (new)

If either of the additional Cat subscriptions is present (Performance + Grade Connectivity or Performance Pro), you will be able to use Remote Access through both VisionLink and Trimble Construction Cloud, when the Primary Cloud Service is set to Trimble Construction Cloud.

The following table shows which online service features are available, depending on the gateway combinations and the Primary Cloud Service that is set:

Gateway Type	Additional Cloud Subscriptions	File Management	Tag Files	Remote Access	Status Message
Primary Cloud Service = None					
None	Performance + Grade Connectivity	N/A	N/A	Cat	N/A
None	Performance Pro	N/A	N/A	Cat	N/A
Primary Cloud Service = Trimble Construction Cloud					
SNM94x radio	-	Trimble	Trimble	Trimble	Trimble
SNM94x radio + PLE702 A6N2 network manager	-	Trimble	Trimble	Trimble	Trimble
PL radio + PLE702 A6N2 network manager	-	Trimble	Trimble	Trimble	Trimble
PL radio + PLE702 A6N2 network manager	Performance + Grade Connectivity	Trimble	Trimble	Trimble + Cat	Trimble
PL radio + PLE702 A6N2 network manager	Performance Pro	Trimble	Trimble	Trimble + Cat	Trimble
Primary Cloud Service = VisionLink					
PL radio + PLE702 A6N2 network manager	Performance + Grade Connectivity	Cat	Cat	Cat	Cat
PL radio + PLE702 A6N2 network manager	Performance Pro	Cat	Cat	Cat	Cat

2.10 Receiving projects and designs from WorksManager

If you are connected to WorksManager, the service can push the latest project and design files to your system. Using WorksManager requires a paid subscription.

When enabled, the grade control system checks WorksManager for updates at start-up and automatically downloads any updates. If there are any subsequent changes to the files, the service sends them to the grade control system. The operator can also manually check for updates by tapping the Refresh button on the Job Setup screen.

NOTE –The version of the project and design on WorksManager is the controlling version, so you must use WorksManager to publish, update, or delete the project or design. If you attempt to modify files on the grade control system, they will be restored to their online state by WorksManager.

Your system can have a combination of local projects and projects from WorksManager. If a project or design is from WorksManager, a cloud icon appears beside it on the Operator App's Job Setup screen. The cloud icon states are:

	For a project: All the project files, including design files, are successfully downloaded. For a design: The design is successfully downloaded from WorksManager and is available to use. The design name will include the version number.
	For a project: Part of the project is still to be downloaded. For a design: The design on WorksManager is new or updated, and is queued to download to your system. It is not currently available to select. If a number appears beside the icon, it represents the design download percentage.
	There was an error downloading the project or design. It is not available to select.
No icon	The design is not from WorksManager.

The download process performs a check to ensure designs are valid.

2.10.1 Troubleshooting

Viewing warnings and errors

To view errors or warnings encountered while connecting to an online service, see the Web Interface's Monitor > Program Log screen.

Viewing the state of published files

You can view the state of published files in WorksManager.

Design validation

If a downloaded file is invalid, the Operator App reports an **Issue >message**.

There is an issue with the file in WorksManager. Check that the file is valid and that the design elements are supported in the grade control system.

For more information on the validation process, see [Design validation](#).

Perform a TCC sync

For new EC520 devices, perform a TCC (Connected Community) sync to authenticate the device before you can use WorksManager.

WorksManager updates project/design that operator is using

If WorksManager updates a project that the operator is currently using, the Operator App reports **Project updated remotely**.

If WorksManager updates a design that the operator is currently using, and auto-archive is enabled, the grade control system deletes the design and downloads the new version. The Operator App reports **Design deleted remotely**.

The dashboard displays so the operator can select the updated project/design. If Autos is engaged, the system waits until Autos is disengaged before returning to the dashboard.

Project or design renamed on WorksManager appears to be deleted

If a project or design is renamed in WorksManager, the grade control system believes that it is deleted. The system deletes the local copy and downloads the new version. The Operator App will report either **Project deleted remotely** or **Design deleted remotely**.

Download connection is poor

If the download connection is poor, causing low download speed, the operator can select the project and design in the Operator App's Job Setup screen. This prioritizes these downloads above other projects and designs in the queue.

Download fails to complete

If a download fails to complete in the allotted time, the partial download can resume when the internet connection is back online.

2.10.2 Known issues

TCC sync delays WorksManager design download

An issue has been reported where the user could not apply a new design, despite download status of 100%.

The grade control system delays installing a design if it is simultaneously syncing large files from TCC. The design becomes available after the TCC sync completes.

2.11 Syncing files with VisionLink Productivity

If you have a configured account, you can sync files between the grade control system and VisionLink Productivity.

NOTE –VisionLink Productivity requires a paid subscription.

The supported file types are:

- Files sent from VisionLink Productivity to the grade control system:
 - Projects and designs (.dsz, .vcl, and .cal files)
- Files sent from the grade control system to VisionLink Productivity:
 - .tag files
 - System snapshots

This process requires that your grade control system is connected to an A6N2 gateway device.

Infield designs are not synced to VisionLink Productivity.

2.12 Using coordinate systems

GNSS systems initially provide spherical coordinates (positions in terms of latitude and longitude). For example, GPS uses a geodetic datum called *World Geodetic System 1984*.

A coordinate system converts these spherical coordinates into the plane-rectangular coordinates (positions in terms of Northings and Eastings) used in ordinary plane surveying. This coordinate transformation is called a projection. Different projection methods introduce different types and magnitudes of distortions.

To correct these distortions in initial projections, you can apply a coordinate adjustment. The adjustments are typically used to modify global coordinates to achieve a best fit with regional coordinates. The coordinate adjustments may be published by a government agency. For example, adjustments can be contained in a grid file, such as a Geoid Grid File (.ggf) or a Datum Grid File (.dgf).

So, the process is:

1. Spherical coordinates available from GNSS
2. Coordinate system converts spherical coordinates to plane rectangular coordinates
3. Local adjustments are applied via grid file

Key coordinate files

The file CoordSystemDatabase.xml is a database that includes published grid files.

The system uses CoordSystemDatabase.xml to generate a proprietary file called a .cal file. The .cal file is a site calibration file that tells the system the coordinate system used for a project, and any grid files for local coordinates.

In previous versions of the software, if you wanted to use a custom grid file, you had to add it to CoordSystemDatabase.xml, regenerate the .cal file, and place the grid file in the \GeoData\ folder.

Selecting a coordinate system

When an operator creates a project, they can select the coordinate system to work with - if you grant permission (this feature is recommended for advanced operators and is disabled by default). You can enable it via the Web Interface's Operation > File Management screen.

They can either:

- Create the .cal file for the project by selecting the correct coordinate system (such as a national reference system), or
- Copy the .cal file from another existing project.

This feature can be useful if they don't have a .cal file, but know the published coordinate system requirements and have the right correction source.

For more information on how an operator can use this feature, refer to the *Project and Design Files Guide*.

Using a grid file that isn't published

In previous versions of the software, CoordSystemDatabase.xml and the .cal file must both list any grid files. If they were not in the .cal file, they wouldn't be usable. These grid files that are referenced in CoordSystemDatabase.xml are called *published*. This made it challenging to add new grid files to the system.

However, you can now also add a grid file to a project without needing to mention it in the .cal file. These grid files are called *independent*. Place the file in the \GeoData\ folder.

Dozer measure-up best practices guide

In this guide:

- ▶ Overview of dozer measure-up
- ▶ Preparing for measure-up
- ▶ Mark measure-up points
- ▶ Position the total station instrument
- ▶ Measure-up
- ▶ Using a USB flash drive to transfer measure-up data in .csv format

3.1 Overview of dozer measure-up

3.1.1 Purpose of this guide

This guide supports the precision measure-up of the dozer blade and GNSS receivers as part of the configuration of a Trimble® Earthworks dozer system.

There are two types of machine blade, which affects the location of measure-up targets:

- Straight blade
- Winged blades (U-blade, SU-blade, carry blade, etc)

This guide covers both blade types.

The measure-up procedure generates the values entered into the Measure-up section of the Web Interface's Configure > Install Assistant screen. Sensor data is taken during this process so it is critical that the machine is in the measure-up position both when the values are taken AND when the values are entered into the Measure-up section.

The measure-up procedure requires the use of a total station instrument to achieve the measurement accuracies needed for accurate positioning.

3.1.2 Prerequisites

Before starting the dozer measure-up:

- Know how to set up and use a total station, for example a Trimble TS/SPS Series total station or a Nikon M Series total station.
- Know how to use SCS900/Siteworks Site Controller software, or the total station's on-board software.

- Install the cab, blade, and GNSS components, and validate the installation.
- Complete the preceding sections of the Web Interface's Configure > Install Assistant screen.
- Review this entire guide. This ensures that you understand the full measure-up workflow.

3.1.3 Measure-up workflow

1. Complete the Sensor Correction Check section of the Install Assistant before starting the Measure-up section.
2. Prepare for the measure-up:
 1. Position the machine.
 2. Mark all measurement points.
 3. Place the measure-up targets on the marked points.
 4. Position the total station.
3. Measure the position of all points.
4. Remove the measure-up targets.

3.1.4 Equipment list

For more information, see [Tools and other supplies](#).

3.1.5 Machine setup

Before proceeding with the measure-up process, check the following:

- The EC520 must be mounted per the installation instructions. If it is mounted in a different position or orientation, the accuracy of the system will be compromised.

If the EC520 is replaced at any time, perform the following steps again:

- The Body section of the Install Assistant
- The Pass Match Adjustment section of the Install Assistant

3.2 Check machine wear

3.2.1 Introduction

This section describes the points of the machine that must be checked and adjusted to provide optimal grade control results. If the machine is loose, the grade control system cannot produce optimal results.

For optimal results, you must maintain on an ongoing basis:

- The machine's optimal working condition
- The accuracy of the machine components

3.2.2 Position the machine

Park on a hard, planar surface as near to level as possible. A workshop concrete floor is an ideal calibration pad.

If needed, put blocks under the cutting edge so the target centerline can line up with the edge of the cutting edge without interfering with the surface underneath.

3.2.3 Check the cutting edge

The cutting edge is defined by points near either end (the recommendation is to measure the cutting edge length 0.3 m/12 inches in from each end), so it is acceptable if the blade has greater wear at one end than the other. However, the cutting edge must form a straight line with a maximum horizontal deviation from a string line of 0 ± 0.005 m (0.2 inches).

If the wear on the blade is inconsistent or the cutting edge is bent (in the forward/back direction), system guidance and cut surfaces will be inaccurate.

3.3 Preparing for measure-up

3.3.1 Overview

A precision measure-up requires:

- A stable, flat pad for the machine
- A stable, safe position for the total station instrument

3.3.2 Set up the machine for measurement

Machine position

Park the machine on a level, smooth, hard surface.

Blade position

Set the blade pitch to the operator's preferred working pitch:

- For C-frame and single tilt machines, position the blade approximately square to the machine.
- For dual tilt machines, set the blade to the mid-pitch position:
 - a. Activate dual tilt mode.
 - b. Raise the blade enough to fully tilt left and right without touching the ground.
 - c. Tilt the LEFT blade tip fully down.
 - d. Tilt the RIGHT blade tip fully down.
 - e. Level the blade so it is approximately parallel with the bottom of the track shoes.
- For all machines:
 - a. Lower the blade to the neutral working position, with the bottom of the cutting edge approximately level by eye with the bottom of the track shoes. Use blocks

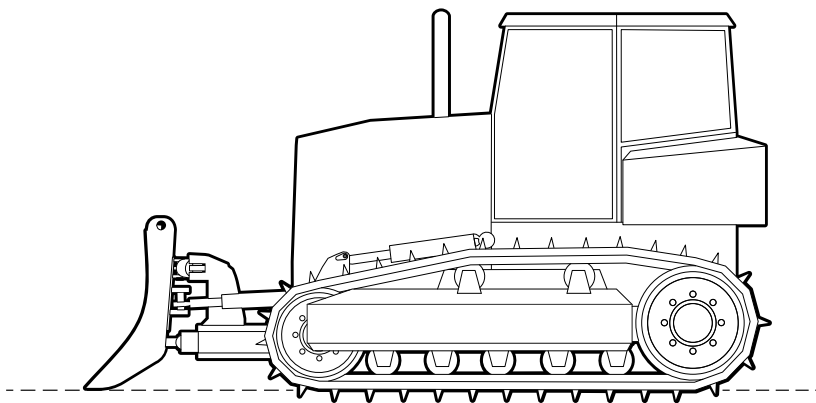
if necessary. Make sure the blade tilt stays the same as the blade comes to rest.

- b. If the blade tilt changes as the blade is lowered:
 - Lift the blade slightly
 - Slightly adjust the blade tilt
 - Lower the blade again to the working position
 - Repeat this step until there is no change in blade tilt when lowered to the working position.

NOTE –For information on switching between dual and single tilt, refer to the machine manual.

NOTE –When configuring an Inactive Body IMU machine, blade rotation is not measured/compensated for on VPAT / PAT / 6-way blades when a single 3D positioning source is being used.

NOTE – Make sure the blade is completely stable and will not drift for the duration of the measurements. For cab-mounted receivers, once the blade is in the measure-up position, do not move it until all measurements, including the C-frame and blade angle measurements, are complete.



3.4 Mark measure-up points

3.4.1 Introduction

This section describes how and where to attach the acrylic targets for the total station to measure.

NOTE –Do not use DR measurement mode. Always measure to acrylic targets to get an accurate measure-up.

3.4.2 Best practices

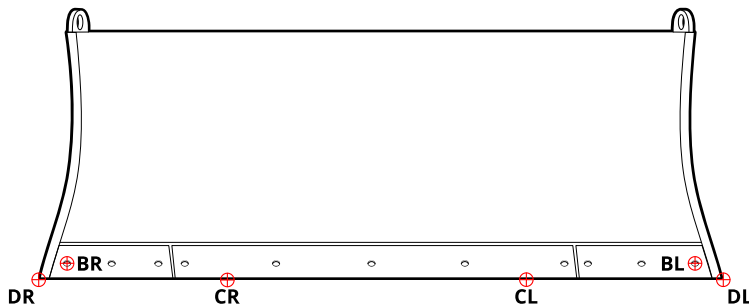
NOTE – Position targets accurately. Your measurements cannot be more accurate than your target positions.

- *Clean* all measurement points.
- *Mark* all measurement points with an acrylic target.
- *Label* all measurement points with the point ID. Write the ID beside the target so you can see it through the instrument.

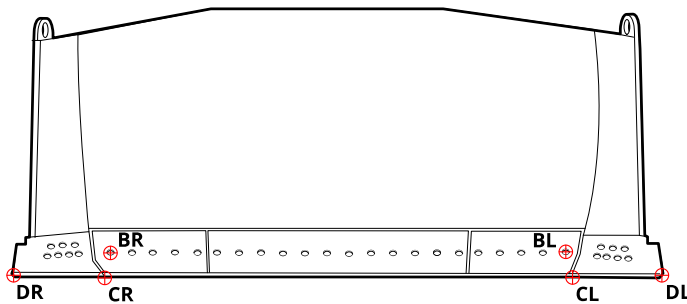
3.4.3 Mark blade measurement points

This section shows how to place the targets for blade measure-up.

Straight blade:



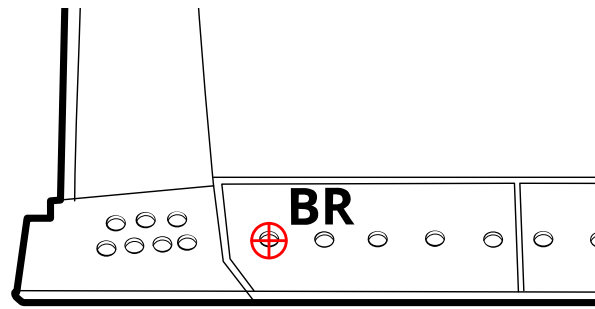
U, SU, or carry blade:



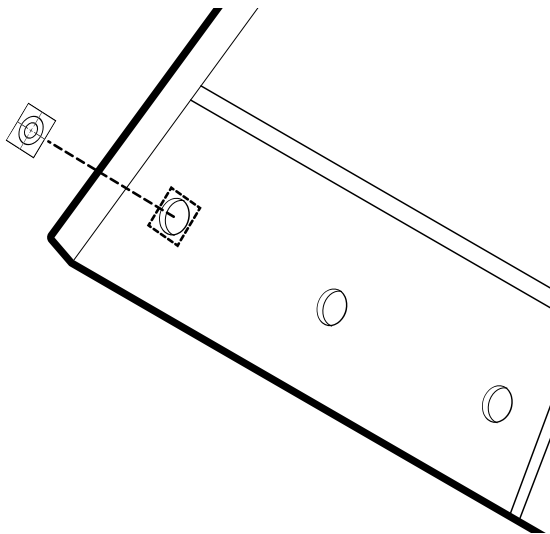
Cutting edge bolts (BR and BL)

1. Locate the outermost bolt head at both the right and left ends of the straight section of the blade. These points should be equidistant from the centerline of the blade.

NOTE –If there are multiple bolt lines on the left and right ends of the blade, place BR and BL on the center section on the line of bolts that connect the cutting edge to the blade.



2. Use a straight edge to accurately mark a cross on the center of each bolt head.
3. Attach acrylic targets so the cross hairs on the targets align with the cross marks on the bolt heads.



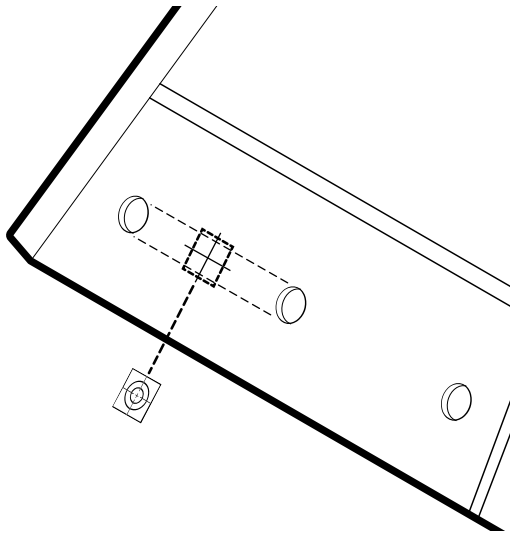
NOTE –This image shows a straight blade. On U, SU or carry blades, do not place the target on the wing section, place the target on the first bolt head on the straight section of the blade.

Cutting edge bolts (BR and BL) – alternative method for recessed bolts

If the bolt heads are recessed, sticking the acrylic target on the recessed bolt head can introduce an error. In that situation, attach the target between the first and second bolt:

1. Use a straight edge to draw a line between the top of the first and second bolt heads that are on the straight blade edge closest to the blade tip.
2. Draw a second line between the bottom of the two bolts' heads.
3. Measure halfway between your two lines and use a straight edge to draw a horizontal line.
4. Measure halfway between the first and second bolt and use a straight edge to draw a vertical line.
5. Place acrylic targets directly over the cross you have drawn. The targets on the left of

the blade and the right of the blade should be equidistant from the centerline of the blade.

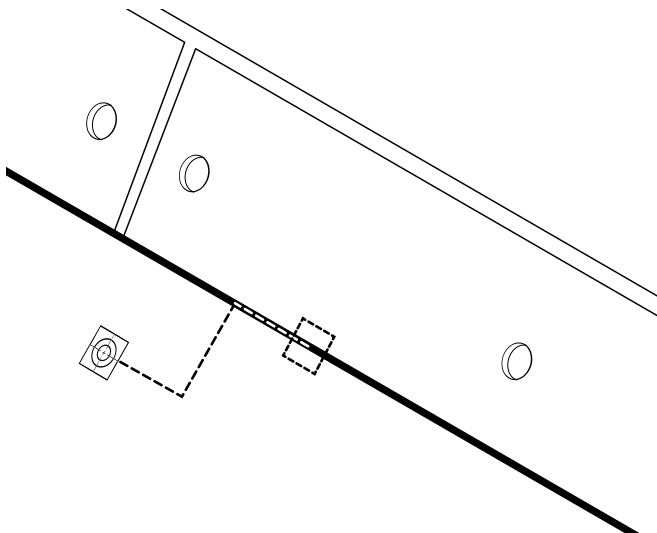


NOTE –This image shows a straight blade. On U, SU or carry blades, do not place the target on the wing section, place the target on the straight section of the blade.

Cutting edge (CR and CL)

Make sure that these points are inset from each end. Do not position them on the wings or a worn section of the cutting edge.

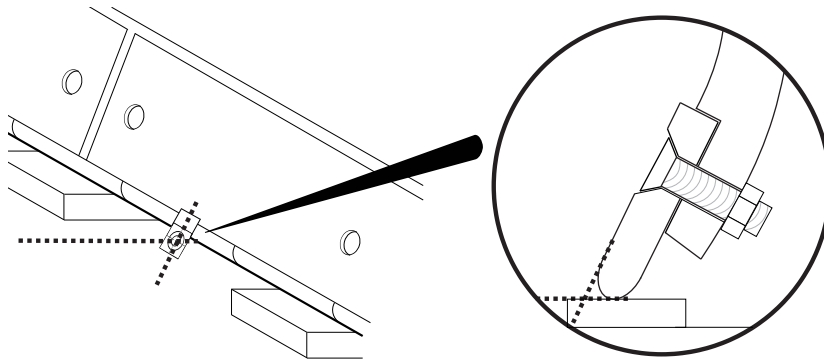
Place the targets half-on and half-off the cutting edge. Where the edge is sloped, place the target half-on and half-off the lowest part of the edge.



NOTE –This image shows a straight blade.

If the cutting edge is curved (either because it is worn or because it is a new, large blade), the horizontal crosshair on CL and CR must be horizontally aligned with the lowest point on the blade. To achieve an accurate measurement, rest the blade on blocks and, if necessary, extend the acrylic target so it can hang over the curved edge.

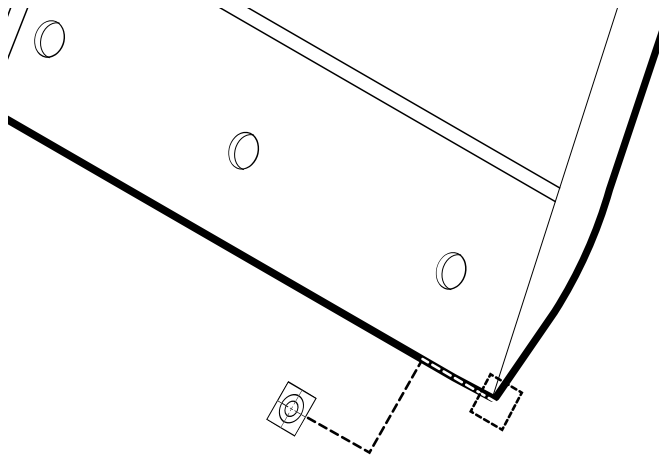
Failure to measure to the lowest point on the cutting edge will reduce accuracy.



Blade tips (DR and DL)

1. Locate the blade extents – the points located on the left and right ends of the widest part of the blade, typically on the cutting edge.
2. Align the center of the target with a blade extent and fix the target to the blade.

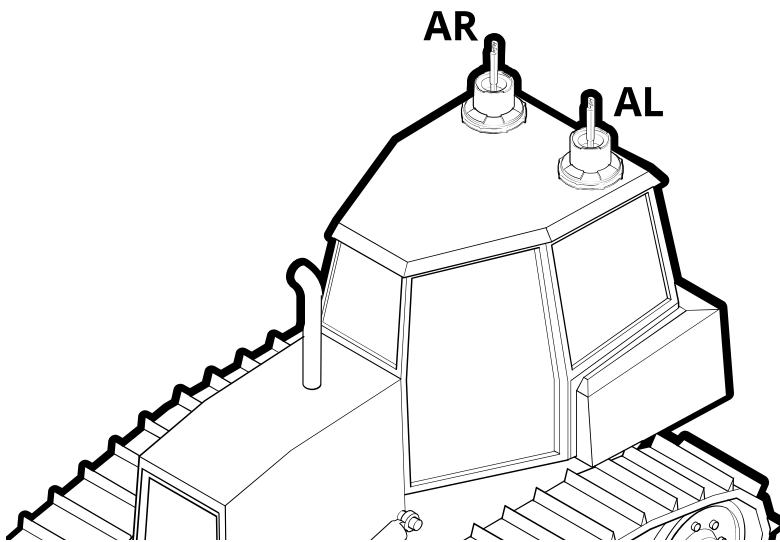
NOTE –This measurement only requires accurate Northings and Eastings. For this reason, the vertical location of the target is not critical.



NOTE –This image shows a straight blade.

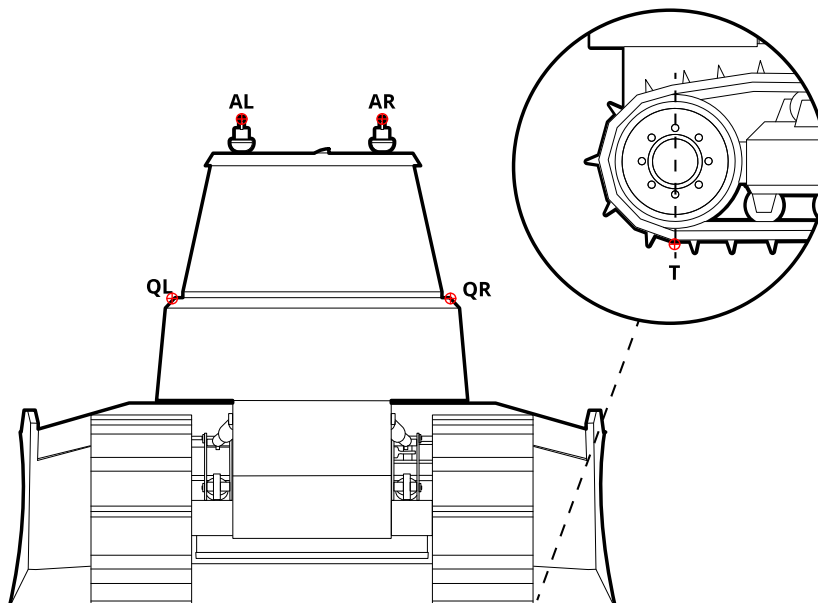
3.4.4 Mark cab-mounted GNSS measurement points (AR, AL, QR, QL and T)

Mark cab-mount GNSS receivers with Active Body IMU sensors (AR and AL)

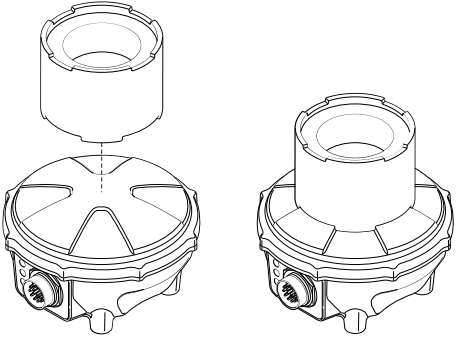
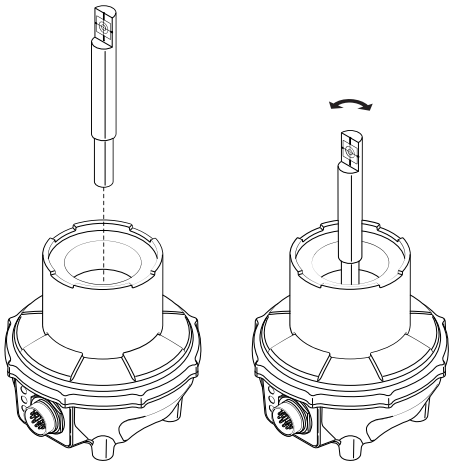


Mark cab-mount GNSS receivers with Inactive Body IMU sensors (AR, AL, QR, QL and T)

NOTE –These steps only apply to cab-mounted GNSS receivers with Inactive Body IMU sensors. For blade mounted receivers, see [3.4.5 Mark blade-mounted mast measurement points \(MX, FX, GX\)](#).

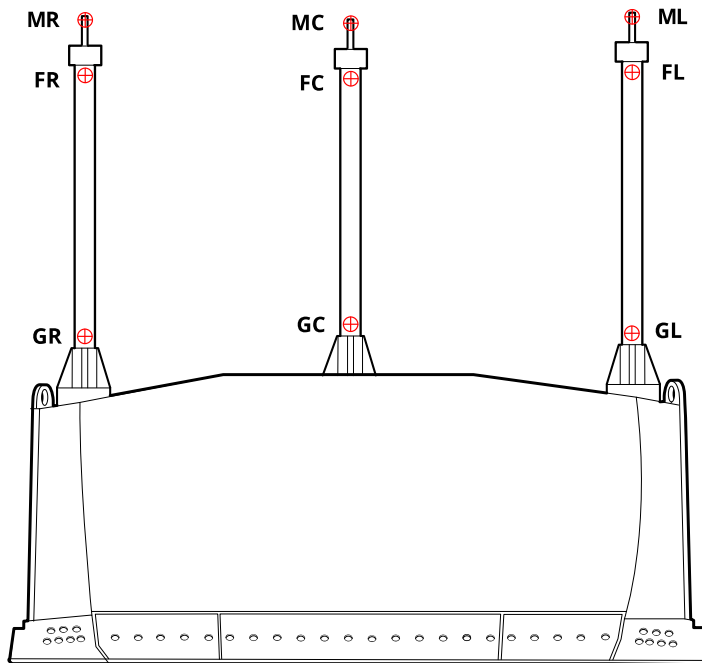


The measurements of the GNSS receivers or antennas are made using the measure-up target kits (P/N 105568).

Steps	Description
	Mount the base on top of the GNSS receiver or antenna radome you are going to measure. The base should fit securely onto the features of the radome. If the base does not engage the radome features securely, flip the base over, and try to mount it again.
	Insert the target rod into the base. Rotate the rod so the acrylic target faces the expected position of the total station.

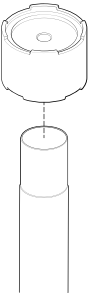
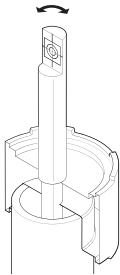
3.4.5 Mark blade-mounted mast measurement points (MX, FX, GX)

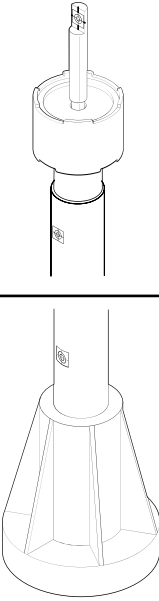
NOTE –These steps only apply to blade-mounted masts. For cab-mounted receivers, see Chapter 3.4.4, Mark cab-mounted GNSS measurement points (AR, AL, QR, QL and T).



You must use a measure-up target kit (P/N 105568) to measure the MR, MC and ML mast points.

Fixed mast

Steps	Description
	Mount the base of the measure-up target on top of the mast so that the side with the narrower diameter hole is on top.
	Insert the target rod into the base. Ensure that the rod rests on the base of the measure-up target and the end of the rod extends into the hollow space inside the mast. Rotate the rod so that the target faces the expected position of the instrument.

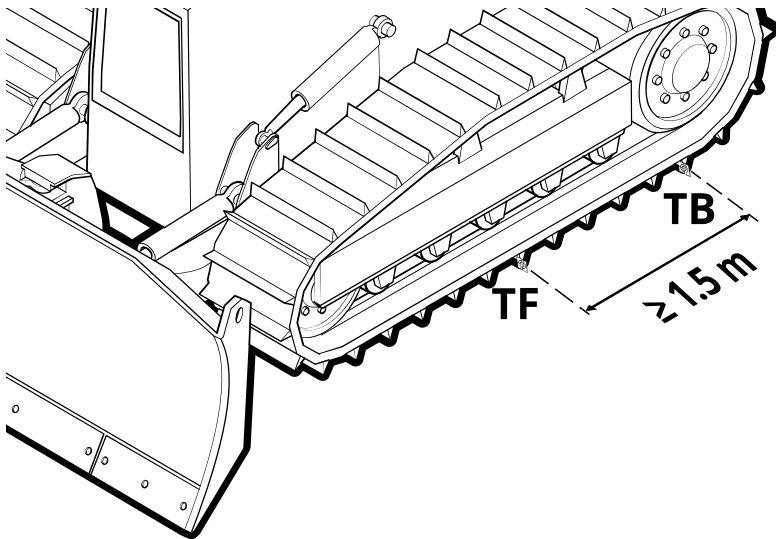
Steps	Description
 The diagram consists of two parts. The top part shows a close-up of a mast's upper section with a vertical rod passing through a circular flange. An acrylic target (FR) is attached to the top of the rod. The bottom part shows a wider view of the mast's lower section, which is a conical structure. An acrylic target (GR) is attached to the bottom of the mast. A horizontal line is drawn across the middle of the diagram, separating the two views.	Attach an acrylic target (FR) to the top of the mast and an acrylic target (GR) to the bottom of the mast. Place FR and GR on the surface of the mast such that a line between FR and GR is parallel to the centerline of the mast.

3.4.6 Mark track edge measurement points (TF and TB)

Accurately place targets on the outer edge of the bottom track, as shown below.

Align the vertical crosshair on the targets with the edge of the vertical edge of the grouser.

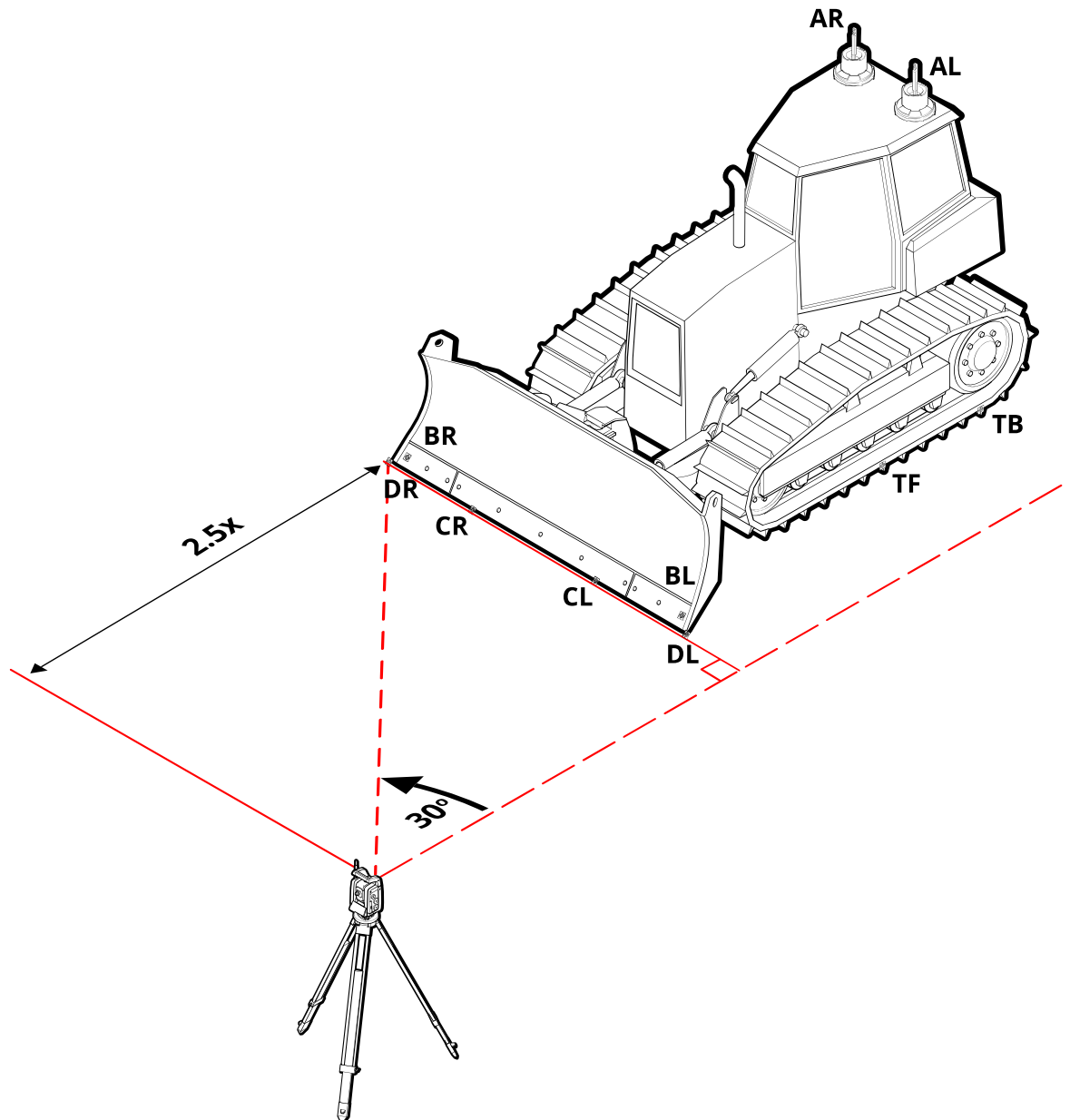
Place the targets with a minimum distance of 1.5 m (5 ft) between them.



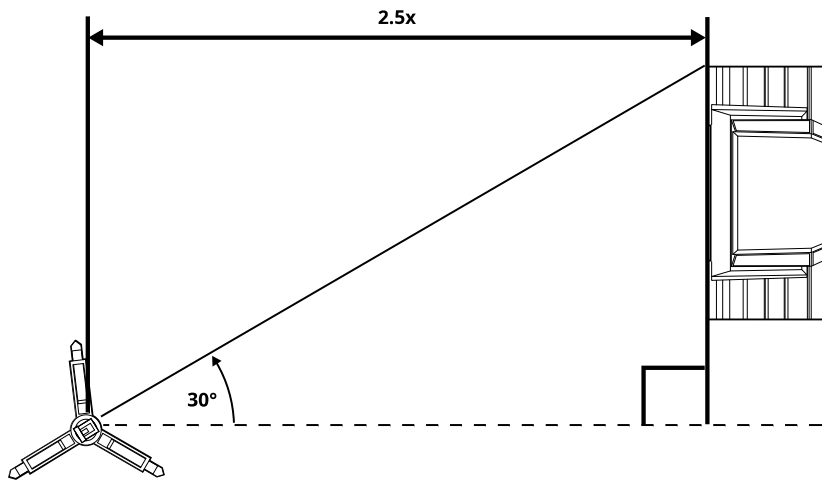
3.5 Position the total station instrument

- Position the instrument to one side of the machine and approximately 2.5 blade lengths from the machine.
- You must be able to see all points through the instrument.
- Ensure that the maximum horizontal angle, or angle of incidence, between any measurement point and the total station is less than 30°.

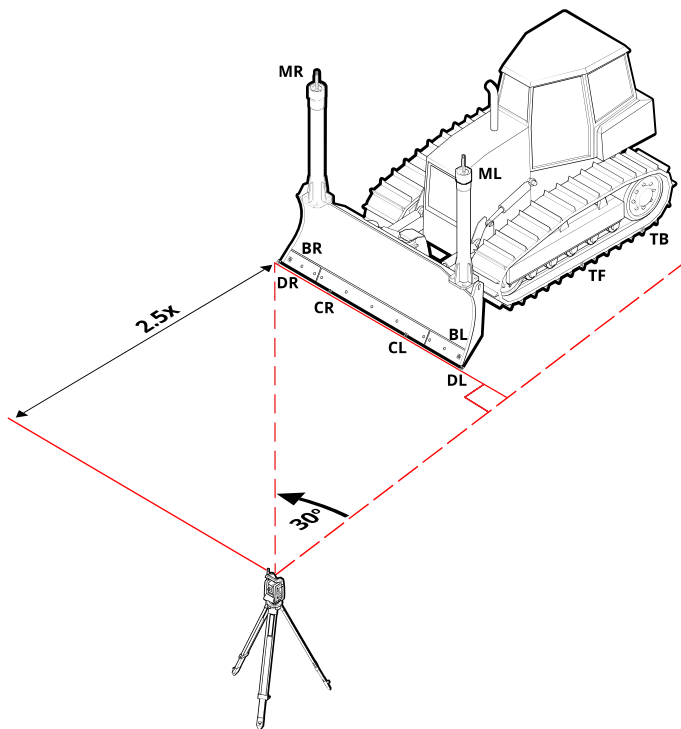
3.5.1 Measure cab-mount GNSS receivers with Active Body IMU sensors



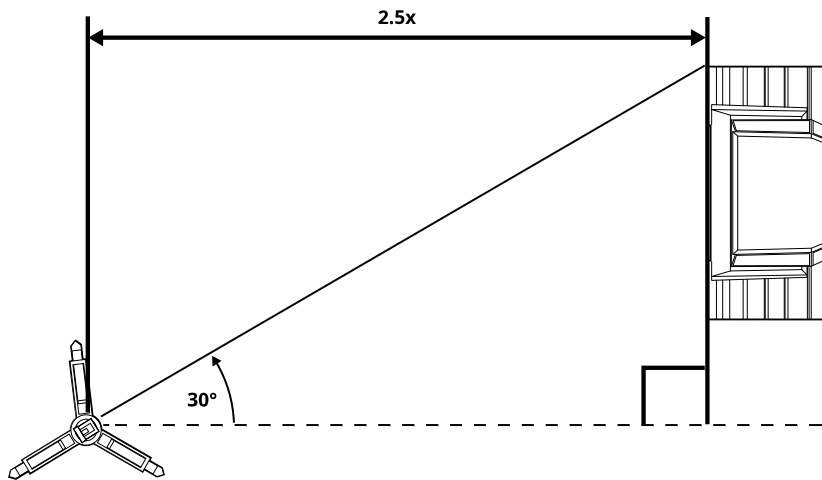
3.5.2 Measure cab-mount GNSS receivers with Inactive Body IMU sensors



3.5.3 Measure blade-mount GNSS receivers



3.5.4 Cab-mount GNSS receivers with Inactive Body IMU sensors



3.6 Measure-up

3.6.1 Overview

This section provides a general outline that enables technicians experienced with Trimble SCS900/Siteworks, or the total station's on-board software, to use a total station instrument to perform a dozer measure-up.

NOTE –Make sure the instrument you are using is within adjustment. Instrument adjustment can be performed on some models with SCS900/Siteworks if required, or by a SITECH[®] service center. An instrument that is not within adjustment may not provide accurate measurement results.

TIP – Use a USB flash drive to transfer measured points from the total station onto the system.

To use this method:

- To enable checking, before you begin the measure-up, make sure that the data collector is set to use the same measurement units and the same coordinate order as set in the Web Interface.
- Name the measured points as they are named in the Measure-up section of the Install Assistant.

For more information, see [Using a USB flash drive to transfer measure-up data in .csv format](#).

3.6.2 Measurement workflow

1. Remove any potential bias from the sensors in the Web Interface's Configure > Install Assistant > Sensor Correction Check.

NOTE – Always complete the Sensor Correction Check before starting the Measure-up section. The Sensor Correction Check removes any potential bias in the sensors, ensuring a successful measure-up.

2. Establish a position for the instrument.

TIP – Use short coordinates to avoid having to enter extra digits in the Measure-up section in the Web Interface. For example, use Northing = 50, Easting = 30, Elevation = 10 for the instrument position.

3. Measure all measurement points.

NOTE – The Target Height for the AR and AL measurements must be *0.00*.

4. Enter the measured coordinate values for each point into the fields in the Enter Coordinates screens in the Web Interface's Configure > Install Assistant > Measure-up section.
5. The calculated Check Lengths values show the slope distance and elevation difference between the measured positions of the targets.

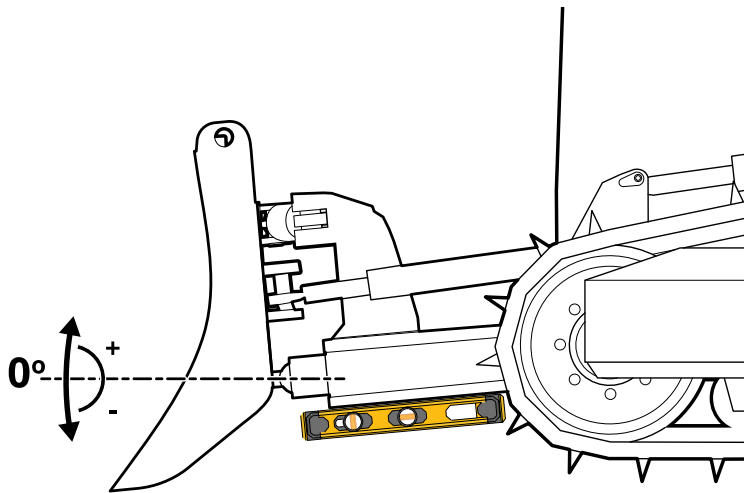
The Check Lengths values enable you to check that the entered positions are approximately correct. Take time to review these as they can indicate data entry errors.

WARNING – Moving a machine with a GNSS measure-up tool attached can dislodge the tool from the machine. Always remove and store removable measure-up tools prior to moving the machine. Only move the machine when instructed in the Measure-up section. When the measured coordinate values are entered, the machine must be in the same position it was in when the measurements were taken.

3.6.3 Measure the C-frame or push arm angle

Ensure that the digital level is parallel with the centerline of the machine.

This image shows a C-frame dozer. On push-arm dozers, the underside of the push-arm can have several different angles. Avoid making the measurement at the blade-end of the push-arm. Instead, place the digital level under the push-arm adjacent to the front idler.

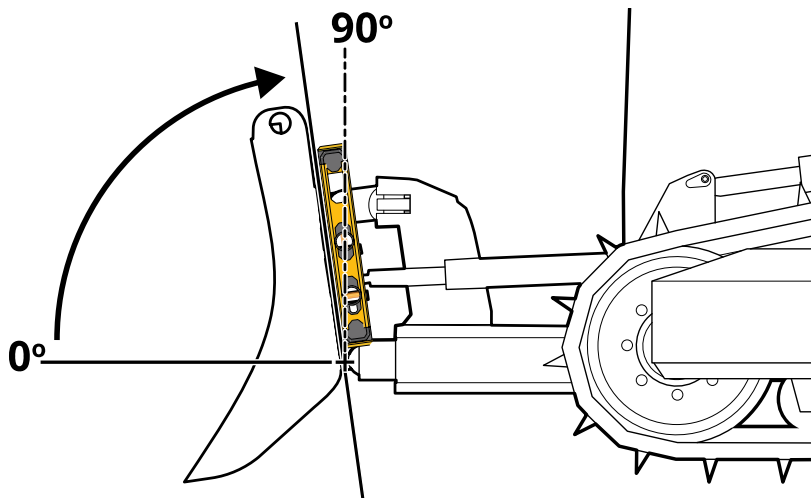


3.6.4 Measure the blade pitch

If the back of the blade is pitched:

- away from the cab - set the Pitch Direction to Forward
- toward the cab - set the Pitch Direction to Back

The image below shows the back of the blade pitched away from the cab.



Measure the blade degree from horizontal to vertical up to 90 degrees. The degree value has the following rules:

- ≤ 90 degrees
- > 0 degrees
- 1 - 69 degrees permitted but with a warning

3.6.5 Take digital level readings for machine mainfall

To take the machine mainfall reading, place a digital level on the insides of both the left and right tracks. Average the two readings and enter the averaged value into the Web Interface.

3.6.6 Entering the values into the Measure-up section in the Web Interface

Consider the following tips:

- Set your coordinate order in the Web Interface's Units dialog first to match your measured point coordinate order in SCS900/Siteworks.
- If sitting in the cab to enter the measured coordinates, lock out the hydraulics to avoid accidentally moving the blade.
- If using your laptop to enter the measured coordinates, you can select E, N, Elv at the same time for each measured point in a CSV file exported from SCS900/Siteworks and paste all three at the same time into the first field for each point in the Web Interface Measure-up sequences. This can increase the accuracy of data entry and decrease the time taken.
- If the measure-up fails on the Save step and the message, "Measure-up calculations failed. Check the measure-up values and readings, then try again." displays, the distance between two measured points is greater than the range specified in the software. Check that all measured values are entered correctly and accurately.

3.7 Using a USB flash drive to transfer measure-up data in .csv format

You can export comma separated value (.csv) files from most common survey data collectors onto a USB flash drive. You can edit .csv files using a text editor, or by opening the .csv file in a product such as Microsoft® Excel.

TIP – For technicians working in regions where it is common for .csv files to use semicolons to separate fields, and commas as decimal fraction markers, then the Web Interface .csv import tool is also able to import these .csv files.

When formatted correctly, as described below, the .csv file can be imported into the Web Interface from the USB flash drive, reducing the amount of manual data entry required during machine commissioning.

3.7.1 Prerequisites

1. The measurements were made using a data collector running software capable of exporting .csv files.
2. The measurement units configured in the data collector are the same as the measurement units used in the Measure-up section of the Install Assistant.
3. The measured points are named as shown in the Measure-up section of the Install Assistant.

3.7.2 Exporting the machine measurements as a .csv file

On the data collector used to make the machine measurements, export the measured points to a USB flash drive in the usual way. If you are using Trimble SCS900 or Trimble Siteworks software on your data collector, make sure you select Include QA Data in the Export dialog, and select Memory Card as the destination.

3.7.3 If required, edit the .csv file

The final .csv file should resemble the example below when it is opened in a spreadsheet editor. This example shows a typical set of measurements.

The column headings must be:

- In English.
- Spelled as shown.
- Capitalized as shown.

NOTE –The order of the columns and point measurements is not important. You can change the coordinate order for your project to suit the region, and you can measure the points in any order.

Point Name	Northing	Easting	Elevation
AL	790.183	638.487	68.931
AR	789.789	639.446	68.943
BL	787.444	636.621	65.677
BR	786.547	639.016	65.719
CL	787.329	636.794	65.604
CR	786.567	638.814	65.637
DL	787.453	636.52	65.626
DR	786.48	639.101	65.668

If you are editing the .csv file in a text editor, then the contents can look like:

```
Point Name,Northing,Easting,Elevation
AL,790.183,638.487,68.931
AR,789.789,639.446,68.943
BL,787.444,636.621,65.677
BR,786.547,639.016,65.719
CL,787.329,636.794,65.604
CR,786.567,638.814,65.637
DL,787.453,636.52,65.626
DR,786.48,639.101,65.668
```

If you are editing a semicolon separated .csv file in a text editor, then the contents can look like:

```
Point Name;Northing;Easting;Elevation
AL;790,183;638,487;68,931
AR;789,789;639,446;68,943
BL;787,444;636,621;65,677
BR;786,547;639,016;65,719
CL;787,329;636,794;65,604
CR;786,567;638,814;65,637
DL;787,453;636,52;65,626
DR;786,48;639,101;65,668
```

When you save the edited file, you must save it as a .csv file.

3.7.4 Import the .csv file into the Web Interface

1. Insert the USB flash drive into the device running the Web Interface.
2. Proceed through the Web Interface's Install Assistant > Measure-up section as usual until you reach the .csv file upload screen.

TIP – If you choose the Skip to Entry option when it is offered, you will still have an opportunity to upload your .csv file.

3. In the .csv file upload screen, browse to the required file.
4. Select the Upload button. The file is uploaded and the measured points in the .csv file are copied into the Web Interface.
5. Proceed through the subsequent screens and confirm that the point data has been correctly imported.

NOTE – After measuring the points, the machine *must not be moved* until the .csv file import completes, and all the screens in the Measure-up section have been completed.

Dozer measure-up and sensor calibration validation guide

In this guide:

- ▶ Overview of dozer measure-up and sensor calibration validation
- ▶ Check static blade guidance on level ground
- ▶ Check static blade guidance on a slope
- ▶ Check dynamic blade guidance on a slope
- ▶ Save and activate machine files

4.1 Overview of dozer measure-up and sensor calibration validation

4.1.1 Purpose of this guide

This guide supports the validation of the dozer measure-up and sensor calibration as part of the commissioning of a new system.

The validation procedures check the accuracy of the:

- Stationary blade guidance on a level surface
- Stationary blade guidance on a sloping surface

The checks must be performed in the above order.

If the required accuracy is not achieved, the procedures describe diagnostic methods for identifying issues.

4.1.2 Prerequisites

Before starting the dozer measure-up and sensor calibration validation:

- Install the cab, GNSS, and blade components, and validate the physical installation.
- Complete all the sections in the Web Interface's Configure > Install Assistant screen.
- Complete or redo the following sections of the Install Assistant for new installs, when the body sensor is changed or when there is a pass matching issue:
 - a. Sensor Correction Check
 - b. Body

- c. Measure-up
- d. Pass Match Adjustment
- In the Operator App's Work Settings > Blade screen, verify the cutting edge lengths are correct.
- Know how to set-up and use a total station or a GNSS rover, and data collector.
- Validate the total station or GNSS rover on a known control point.
- If you are using a GNSS rover, make sure both the Earthworks system and the rover are using the same site calibration.
- A site with:
 - A level pad
 - A sloping pad with a cross slope representative of the slope the machine will work on
- Review this entire guide. This ensures that you understand the full validation workflow.

4.1.3 Equipment list

For more information, see [Tools and other supplies](#).

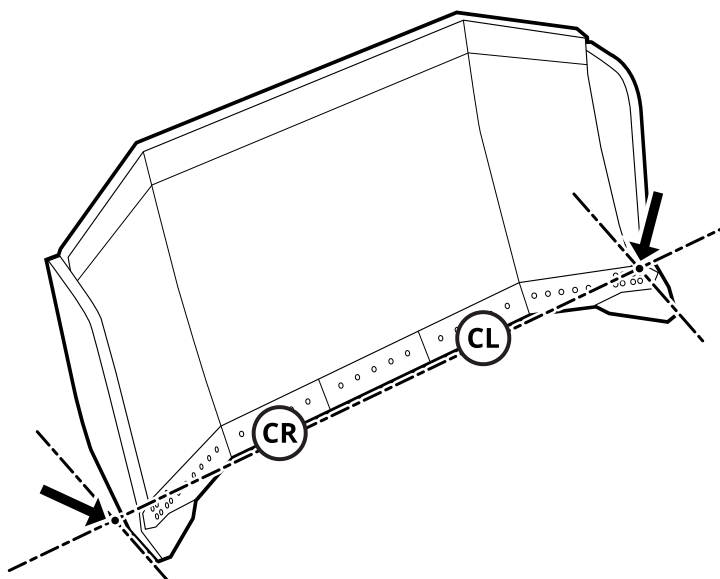
4.2 Check static blade guidance on level ground

4.2.1 Check - Active Body IMU machines

NOTE – On Active Body IMU systems, guidance is only accurate when the machine is moving, or if the machine is stationary, when the blade has been raised and lowered.

1. Move the machine on to a level pad.
2. Square and level the blade slope.
3. Raise the blade fully and then lower the blade onto the ground to initialize the sensors.
4. Survey the positions of the blade tips, and compare the surveyed positions with the reported positions in the Web Interface's Monitor > Machine Positions screen. Note the two sets of positions.

NOTE – On blades with angled tips, survey the position of the line projected from CL and CR.



5. Rotate the blade so that the right tip is fully
 - a. back – repeat steps 3 and 4
 - b. forward – repeat steps 3 and 4

4.2.2 Check – Inactive Body IMU machines

1. Square the blade and put in the operator's preferred working position.
2. Move the machine on to a level pad:
 - a. Move the machine at least one machine length forward in a straight line.
 - b. Ensure no orange operational state messages are displayed.
3. Survey the positions of the blade tips, and compare the surveyed positions with the reported positions in the Web Interface's Monitor > Machine Positions screen. Note the two sets of positions.

4.2.3 Corrective action

The surveyed positions of the blade tips should be within 0.02 - 0.03 m (~1 inch) of the reported positions. If not, repeat the Measure-up section of the Web Interface's Configure > Install Assistant screen.

4.3 Check static blade guidance on a slope

NOTE – On Active Body IMU systems, guidance is only accurate when the machine is moving, or if the machine is stationary, when the blade has been raised and lowered.

4.3.1 Prerequisite

You must have performed the procedure, and any necessary corrective actions, described in [4.2 Check static blade guidance on level ground](#).

4.3.2 Check

1. Move the machine on to the sloping pad so that the machine body has a slope across it.
2. Align the blade so it is parallel to the slope.
3. On Active Body IMU systems, raise the blade fully and then lower the blade onto the ground to initialize the sensors.

NOTE –Do not raise and lower the blade to initialize the sensors on Inactive Body IMU systems.

4. Survey the positions of the blade tips, and compare the surveyed positions with the reported positions in the Web Interface's Monitor > Machine Positions screen. Note the two sets of positions.
5. Rotate the blade so that the right tip is fully back and repeat steps 3 and 4.
6. Rotate the blade so that the right tip is fully forward and repeat steps 3 and 4.

4.3.3 Corrective action

The surveyed positions of the blade tips should be within 0.025 m (1 inch) of the reported positions. If not, repeat the Pass Match Adjustment section of the Web Interface's Configure > Install Assistant screen.

4.4 Check dynamic blade guidance on a slope

4.4.1 Complete the Sensor Correction Check

For all new installs, when the body sensor is changed or when there are pass matching issues, complete or redo the following sections of the Install Assistant in the order listed:

1. Sensor Correction Check
2. Body
3. Measure-up
4. Pass Match Adjustment

NOTE –In version 2.23 and later, you can run the Sensor Correction Check after the measure-up process. If the check is within 100 operating hours of the measure-up occurring, the system will update the measure-up using the corrected blade sensor values.

4.4.2 Check

1. The machine Pass Matches on level ground.
2. The machine Pass Matches on a slope.
3. Passes match when made in
 - a. The same direction
 - b. Different directions

4.4.3 Corrective action

When the machine is working in Autos, determine if:

- The issue is constant, determine if the constant issue occurs when working on:
 - Level ground and slopes - redo the Body and Measure-up sections
 - Slopes only - redo the Pass Match Adjustment section
- The issue is not constant, especially when the blade drifts off grade while making a pass and drifts back to grade when the machine is stopped - redo the Sensor Correction Check section. The Sensor Correction Check requires machine access to a level surface and a slope of at least 10%.

4.5 Save and activate machine files

4.5.1 Save a machine file

The current settings save automatically in the system memory and persist over power cycles. Current settings are overwritten when you activate or restore a saved machine file. Saving the current settings to a machine file makes the settings available for activating or restoring as needed.

NOTE –Users with Operator Plus accounts can access the Machine Settings Files screen to apply machine files. From within the Operator App, tap System Settings > Manage Machine File. To give an operator Operator Plus access, within the Web Interface open Configure > Install Assistant > User Permissions.

Consider when to save a machine file:

- After installing, commissioning and validating guidance accuracy of the system
- Before and after a new valve calibration so you can compare previous settings to the new settings
- After completing and validating a measure-up (delete the previous machine file to avoid using the old settings instead of the new settings)
- After completing a new machine setup. You must complete a new valve air purge followed by a valve calibration and any other required machine calibrations for each machine using additional attachments and sensors. The machine file is then available for activation when the machine setup changes.

To save a machine file:

1. Open the Machine Settings Files page in the Web Interface.
2. Select  and then Save Settings.
3. Give the file a unique name and save.

4.5.2 Activate, or restore, a machine file

Activating, or restoring, a machine file is useful for returning the system back to a state that was previously saved.

4 Dozer measure-up and sensor calibration validation guide

Most system settings are overwritten with the values from the machine file when that machine file is activated or restored.

After activating a machine file, check the following items to ensure accurate guidance:

- Cutting edge wear
- Currently loaded project and 2D bench or 3D design settings
- Operational settings such as vertical guidance, focus and overcut protection options
- Items such as Wi-Fi configurations and delayed shutdowns do not reset when activating a machine file.
- System licenses are not affected by activating a machine settings file.

To activate a machine setting file:

1. Open the Web Interface and then Advanced > Machine Settings.
2. Select the file to activate.
3. Select Make Active.

Dozer valve calibration setup guide – Inactive Body IMU machines

In this guide:

- ▶ Overview of dozer valve calibration setup
- ▶ Measure the blade arm pivot length – Inactive body IMU machines only
- ▶ Set the blade position
- ▶ Mount a sensor for valve calibration

5.1 Overview of dozer valve calibration setup

This guide supports the setup required for valve calibration on the following machines:

- Inactive Body
- IMU Tilt valve
- C-frame

5.2 Measure the blade arm pivot length – Inactive body IMU machines only

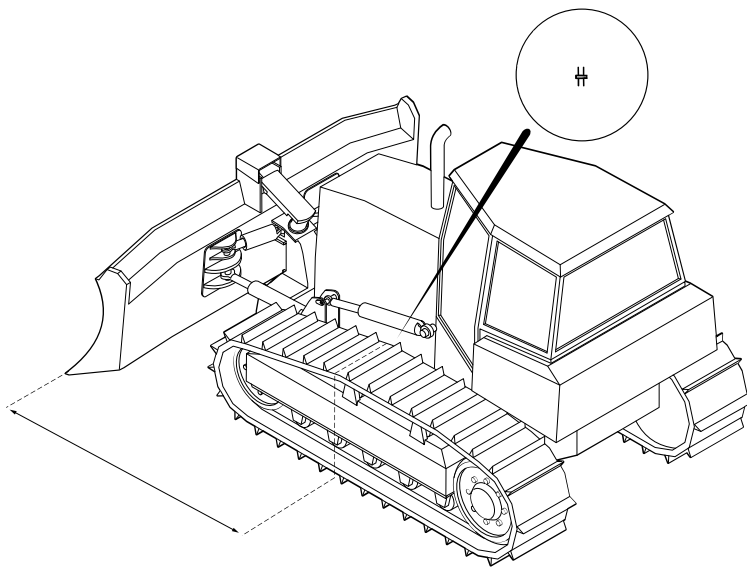
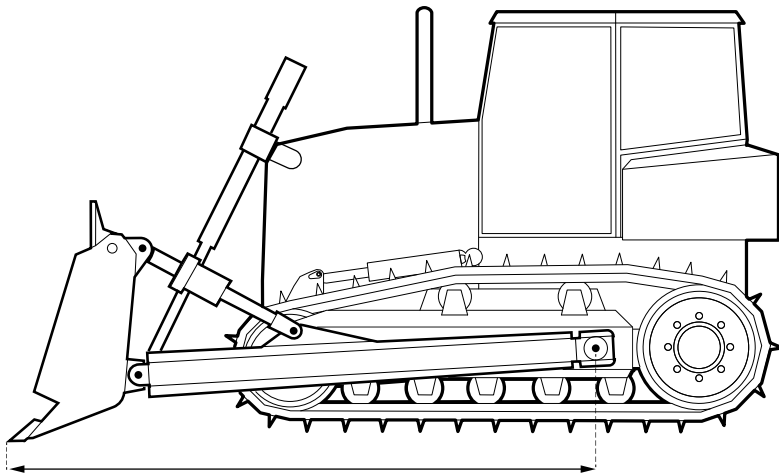
Measure the length of the blade pivot arm. You will need a marker, a straight edge, and a steel tape measure.

1. Put the blade on the ground.
2. Locate the point on the blade arm at which it pivots about the main frame assembly.
3. Project this point outwards, along the tracks and down to level ground.

NOTE – Depending on the machine type, the pivot point may be clearly visible from the side of the machine or may be hidden behind the undercarriage.

TIP – Use a feature on the tracks that aligns with the pivot point on the blade arm when projecting the pivot point to the ground.

4. Use a tape measure to measure the length from the pivot point to the cutting edge of the blade.



5.3 Set the blade position

- For C-frame and single tilt machines, position the blade approximately square to the machine.
- For dual tilt machines, set the blade to the mid-pitch position:
 - a. Activate dual tilt mode.
 - b. Raise the blade enough to fully tilt left and right without touching the ground.
 - c. Tilt the LEFT blade tip fully down.
 - d. Tilt the RIGHT blade tip fully down.

- e. Level the blade so it is approximately parallel with the bottom of the track shoes.
- For all machines:
 - a. Lower the blade to the neutral working position, with the bottom of the cutting edge approximately level by eye with the bottom of the track shoes. Use blocks if necessary. Make sure the blade tilt stays the same as the blade comes to rest.
 - b. If the blade tilt changes as the blade is lowered:
 - Lift the blade slightly
 - Slightly adjust the blade tilt
 - Lower the blade again to the working position
 - Repeat this step until there is no change in blade tilt when lowered to the working position.

NOTE –For information on switching between dual and single tilt, refer to the machine manual.

NOTE –When configuring an Inactive Body IMU machine, blade rotation is not measured/compensated for on VPAT / PAT / 6-way blades when a single 3D positioning source is being used.

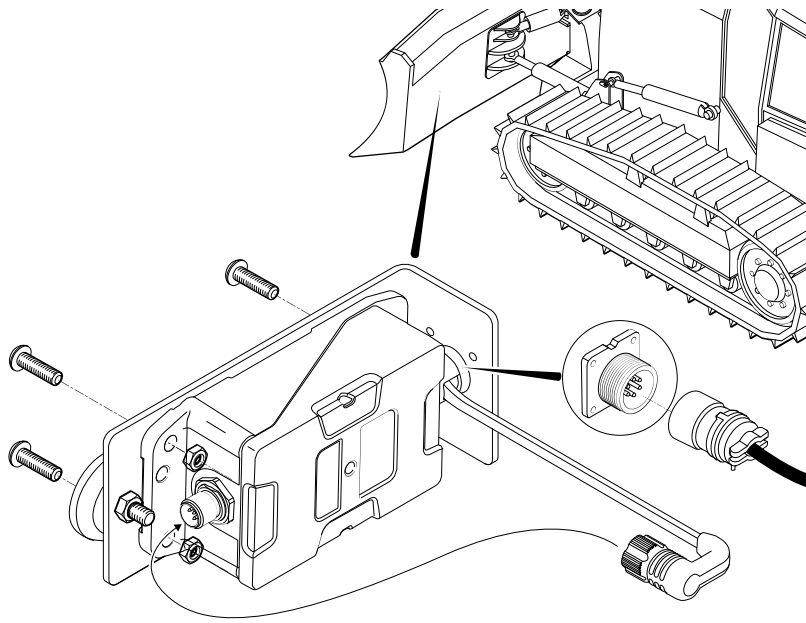
Perform the calibration in single tilt mode.

5.4 Mount a sensor for valve calibration

Mount a temporary GS520 grade sensor on the back of the blade to calibrate the valves.

1. Move the machine on to a level pad.
2. Put the blade on the ground.
3. Approximately square the blade.
4. Attach the grade sensor to the magnetic mount (P/N 194170-00).
5. Attach the magnetic mount and sensor assembly to the back of the blade, on the left side, with:
 - the connector on the sensor facing left
 - the quick disconnect cable connector facing right
 - the long side of the sensor parallel to the cutting edge bolt line
6. Connect the sensor assembly to the quick disconnect on the machine.
7. Follow the instructions in the Valve Calibration section of the Web Interface's Configure > Install Assistant screen to complete the calibration of the valves.

5 Dozer valve calibration setup guide – Inactive Body IMU machines



Dozer Autos tuning guide

In this guide:

- ▶ Overview of valve tuning
- ▶ Tune the lift valves
- ▶ Tune the tilt valves
- ▶ Save Settings

6.1 Overview of valve tuning

6.1.1 Purpose of this guide

This guide describes how to complete a full tune of the Trimble Earthworks automatic system on a supported dozer model.

6.1.2 Overview of tuning

NOTE – Think carefully before performing a full tune of the automatic control system. If you selected a specific machine model during system configuration, then the system is already tuned. In this case, first cut a test surface, and if the surface defects are minor, follow the tuning procedure in the Operator's *Using Autos* guide.

The tuning procedures described in this guide are intended to be completed by a technician and are not part of an operator's day to day work. Tuning may take up to two hours, depending on machine type.

During the tuning procedure it is important that the only changes to the automatic controls performance are due to the valve speed settings. Pass speed, engine power, material type, consistency, water content and blade loading must remain the same from test pass to test pass. To maintain material consistency, you may have to rip and water the test surface after several passes.

TIP – When you complete lift and tilt tuning, then save the machine file. The tuning settings are saved, and are restored when the machine file is restored.

6.1.3 Prerequisites

Before starting a dozer valve tune:

- Complete the installation of all components, and validate the hardware installation.
- Complete the validation of the measure-up and sensor calibration. For more information, see [Chapter 4, Dozer measure-up and sensor calibration validation guide](#).
- Understand how the dozer automatic system operates. Refer to these guides:
 - *Using Autos*
 - *Best Practices When Using Autos*

Find these guides in the Operator App or online.

- In the office or using the Depth and Slope mode in the Operator App, create a level pad design to act as a test design.
- Using the level pad design, create a site approximately 80 m (260 ft) long and 12 m (40 ft) wide. The recommended material is a homogenous mix of loose, crushed gravel and fines with a maximum size of approximately 20 mm (0.8 inches).
- Review this entire guide. This will ensure that you understand the full tuning workflow.

6.2 Tune the lift valves

WARNING –The cutting edge may move abruptly during automatic control system optimization. To avoid injury, when the machine is stationary engage the machine's park brake (if present) and have an operator in the machine cab to supervise the optimization. Maintain adequate clearance from people and objects during the automatic control system optimization.

The goal is to tune the machine so that it can complete a pass at operating speed to the operator's satisfaction. If the cut/fill values on the text ribbon change by less than 0.02 m during a tuning pass, then the system is performing satisfactorily. If the cut/fill values on the text ribbon are changing by more than 0.02 m during a tuning pass, continue tuning the lift valves.

The tuning process involves cutting a series of test surfaces. After each pass, you observe the test surface and then adjust the tuning settings. The typical adjustments are shown in the table below, but some steps require you to adjust the settings in a different direction.

Surface defect	Action
Short, choppy waves	Decrease the Raise Speed, Lower Speeds Decrease the Stability setting as a last option

Surface defect	Action
Smooth	Raise Speed, Lower Speeds and Stability setting are approximately correct
Long, rolling waves	Increase the Raise Speed, Lower Speeds Increase the Stability setting as a last option

6.2.1 Prepare for tuning

Software

In the Web Interface:

1. Open Configure > Valve Tuning.
2. Turn on Temporarily Enable Advanced Operator Tuning.
3. Release the lock on the Web Interface.

In the Operator App:

1. Open System Settings > Configuration > Units to set the units to meters. This allows you to tune to higher accuracy.
2. Edit the text ribbon so that only Cut/Fill Center (Cut center) displays. This allows you to concentrate on the cut/fill values.
3. Load the test design.
4. Open Work Settings > Blade > Vertical Guidance > Center and Select to set the vertical guidance point to the center of the blade.
5. Open Work Settings > Auto Mode > Advanced Options. The valve tuning sliders are on this screen.
6. Record the current values of the sliders.
7. Tap Reset to set all sliders to 100. This sets the tuning values to their defaults.
8. Tap Apply.

Machine

1. On the machine, turn off Eco Mode, if available, and set the throttle to maximum.

NOTE – The throttle should always be set to maximum when using Autos.

2. Make the blade square and level.
3. Make several passes in Autos to warm the hydraulic oil to the machine's normal operating temperature.

6.2.2 Set the unloaded raise and lower speeds

The goal is to determine the optimum tuning settings for the automatic control of an unloaded blade. The procedure also determines the difference between the Raise Speed and Lower Speed settings required to automatically raise and lower an unloaded blade at the same speed.

1. Move the machine onto the design.
2. Move the blade onto grade and engage Autos.
3. Increment the Elevation Offset so that the blade is on-grade and roughly 0.50 m in the air.
4. Use the dozer manual controls to raise or lower the blade about 0.20 m off grade, and then allow Autos to return the blade to grade. Observe the speed at which the blade returns to grade.
5. In the Work Settings > Auto Mode > Advanced Options pane adjust one or both of the Raise and Lower Speed sliders, tap Apply, re-engage Autos, and repeat step 4 until the blade raises and lowers to grade at the same speed. The blade should move quickly and smoothly to grade, with an overshoot of less than 0.01 m. If the blade does overshoot grade, the Autos should only make one correction to reach grade.

TIP – Adjust the slider values by approximately 10 units at a time.

6. Make a note of the settings.

6.2.3 Perform a test pass

1. Set the Elevation Offset to achieve a cut of 0.02 m.
2. Put the blade on grade.
3. Begin the pass and engage Autos. Make the pass at the operator's normal operating speed. The blade should be carrying approximately 25% to 50% of a full load. A 30% load is best.
4. Complete the pass maintaining a constant ground speed.

NOTE – All subsequent test passes must try and replicate the speed, throttle setting, and blade loading of the first pass.

6.2.4 Find the performance transition speeds

The goal is to find, and record, the two sets of tuning settings where a series of passes produce test surfaces that transition between:

- Short choppy waves and a smooth surface
- Long rolling waves and a smooth surface

NOTE – Always try to adjust the Raise Speed and Lower Speed settings by the same amount, and in the same direction. It is important to maintain the relationship between the speeds (that is, the gap between the settings) established in [6.2.2 Set the unloaded raise and lower speeds](#). In some cases, once the blade is loaded with material, the Raise Speed or Lower Speed relationship (that is, the gap between the settings) may need slight readjustment. If adjusting the Raise Speed and Lower Speed does not improve performance, adjust the Stability as the last option.

1. Observe the test surface. If the test surface shows:
 - Short, choppy waves, decrease the Raise Speed and Lower Speed settings.
 - Long, rolling waves, increase the Raise Speed and Lower Speed settings.
 - A smooth surface, increase the Raise Speed and Lower Speed settings.

TIP – Make adjustments to the slider values of approximately 10 units at a time.

2. Perform another test pass as described in [6.2.3 Perform a test pass](#), and adjust the Raise Speed, Lower Speeds and Stability settings as described above.
3. Repeat steps 1 and 2 until one the following sets of transitions are observed:
 - The test surfaces for a series of passes change from choppy to smooth, and then smooth to rolling.
 - The test surfaces for a series of passes change from rolling to smooth, and then smooth to choppy.
 - The test surfaces for a series of passes change from smooth to choppy.

If you started with a smooth test surface, once you have found the Raise and Lower Speed settings where the test surfaces change from smooth to choppy, reset the Raise Speed, Lower Speed and the Stability settings to the values established in [Set the unloaded raise and lower speeds](#), and then decrease the settings until the test surfaces change from smooth to rolling.

Note the settings for each transition.

6.2.5 Tune the valves

NOTE – Always try and adjust the Raise Speed and Lower Speed settings by the same amount, and in the same direction. It is important to maintain the relationship between the speeds established in [6.2.2 Set the unloaded raise and lower speeds](#). In some cases, once the blade is loaded with material, the Raise or Lower Speed may need slight readjustment. When you are satisfied with the Raise Speed and Lower Speed settings, adjust the Stability setting if needed.

1. Set the Raise Speed, Lower Speeds and Stability setting to the values for the choppy to smooth transition found in [6.2.4 Find the performance transition speeds](#).
2. Perform a test pass as described in [6.2.3 Perform a test pass](#), and adjust the Raise Speed, Lower Speed, and Stability setting as described in [6.2.4 Find the performance transition speeds](#). This should mean **lowering** the Raise Speed, Lower Speeds, and Stability setting.

TIP – Make adjustments of approximately 1 unit at a time.

3. Repeat steps 1 and 2 until the test surface just begins to transition from smooth to rolling. This means you will tune through the optimum settings.
4. Repeat step 2 but **increasing** the Raise Speed, Lower Speeds, and Stability setting until the test surface is reliably smooth. Apply these settings.

Lift valve tuning is complete.

6.3 Tune the tilt valves

WARNING –The cutting edge may move abruptly during automatic control system optimization. To avoid injury, when the machine is stationary engage the machine's park brake (if present) and have an operator in the machine cab to supervise the optimization. Maintain adequate clearance from people and objects during the automatic control system optimization.

6.3.1 Overview

It is highly unlikely that you will need to tune the tilt speed, as the default setting normally provides good performance. If you do need to tune the tilt speed, perform the tuning procedure with the blade in the air.

6.3.2 Set the tilt speed

1. Move the machine onto design.
2. Move the blade onto grade and engage Autos.
3. Increment the Elevation Offset so that the blade is on-grade and roughly 0.50 m in the air.
4. Configure the left and right cut/fill text items.
5. Use the dozer manual controls to tilt a blade tip about 0.20 m off grade, and then allow Autos to return the blade to grade. Observe the Autos tilt speed.
6. In the Work Settings > Auto Mode > Advanced Options pane adjust the Tilt Speed slider, tap Apply, re-engage Autos, and repeat step 4 until the blade moves quickly and smoothly to grade, with an overshoot of less than 0.01 m. If the blade does overshoot grade, the Autos should only make one correction to reach grade.

TIP – Make adjustments of approximately 10 units at a time.

7. Apply these settings.

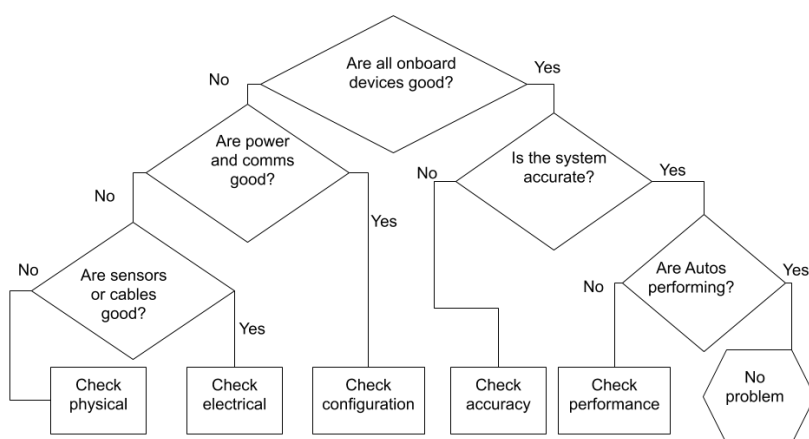
Tilt tuning is complete.

6.4 Save Settings

Save the tuned settings by saving a machine file in the Web Interface.

1. Open the Web Interface's Advanced > Machine Settings Files screen.
2. Select the machine you just finished tuning and select Upload.
3. Save the file with a unique name.

Troubleshooting decision tree



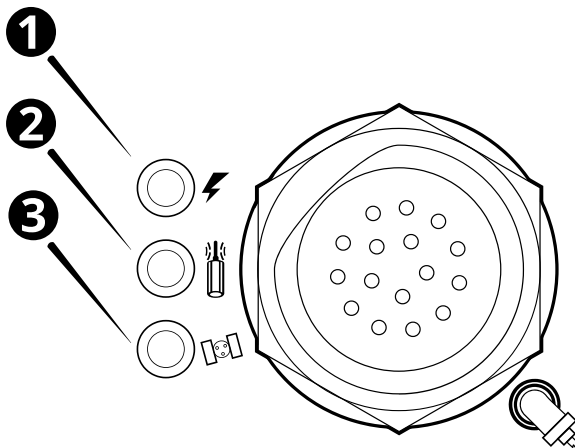
7.1 Check physical

Possible issue	Possible issue and resolution
TD5x0 does not turn on	Cables not connected. Check that the power cable between the TD5x0 and the power source is connected.
	Cable is connected but broken. Check that the power cable between the TD5x0 and the power source is intact and does not have any breaks.

Possible issue	Possible issue and resolution
The right GNSS data link LED is flashing 1 Hz and solid on the left GNSS data link LED and the machine is facing the wrong direction in the guidance views	The left receiver is plugged into the right side cable and the right receiver is plugged into the left side cable. Locate the GNSS splitter cable (P/N 150750-01) and swap the eight pin connectors to the correct sides.
The display shows GNSS Error, No GNSS Position	The GNSS receivers are not connected, check cables and connections or swap the display cable with another cable.
In the Web Interface, when measuring the attachment position, I cannot start the wizard because the GNSS errors are too large or do not have an RTK position.	The GNSS receiver does not receive corrections. Try one of the following: • Ensure you used the correct GNSS measure-up cable (P/N 150775-46). Using incorrect cables or extension cables may increase resistance of the CAN bus. • Note the resistance on the CAN bus and compare to working systems. • Ensure the measure-up cable is attached in the correct location as specified in the instructions on the screen.
On the TD5x0, the wrong action happens when I touch the screen.	Try one of the following: • Be more accurate with your tap gestures. • Activate glove mode, which increases the screen sensitivity. In the TD5x0, open the Settings > Touch > Touch Sensitivity and select Glove.

The MS9xx GNSS receivers have three LEDs next to the harness connector. How these LEDs behave indicates the receiver status, as shown in the following figure.

In a dual GNSS system the left receiver data link should flash at 1 Hz and the right receiver should be solid.



LED	Left	Right
❶ Power	Off – No power On – Power *	Off – No power On – Power *
❷ Data link	Off – No corrections 1 Hz – Corrections [†]	Off – No corrections On – Corrections received from the left receiver [†]
❸ Satellite	Off – Not tracking 2 Hz – Tracking fewer than 4 satellites 1 Hz – Tracking more than 4 satellites *	Off – Not tracking 2 Hz – Tracking fewer than 4 satellites 1 Hz – Tracking more than 4 satellites *

* required

[†] in a dual GNSS system the left receiver data link should flash at 1 Hz and the right receiver should be on (solid)

7.2 Check electrical

7.2.1 TD5x0 display does not turn on

Possible issue	Possible issue and resolution
The display or other devices intermittently powers on and off.	The earth connection may be bad. Test the voltage while the system is off. Then continue to test the voltage and power cycle the whole system to test the voltage under full load.

Possible issue	Possible issue and resolution
When the machine ignition switch is turned on, the display does not turn on.	The boot configuration is not set to turn on when power is applied. Press the black power button on the display. If the display • turns on - consider changing the boot configuration setting in the display system menu. • does not turn on - continue to further tests.

7.2.2 Further tests

1. Check any components with power LEDs (i.e. a GNSS receiver) that the power LED turns on when the ignition switch turns on. Lightbars, if used, should display a short start up pattern when the ignition switch is turned on. If component power LEDs or lightbars:
 - a. Turn on, but the TD5x0 does not turn on, the display cabling or the TD5x0 may be faulty.
 - b. Does not turn on, the power relay or power cabling to the system may be faulty, perform the next step.
2. Measure the voltage between pins D and E on the circular eight pin display cable connector and pins 1 and 4 on the four pin connector on the display cable. If power is:
 - a. Detected – the display may be faulty, replace with a working display.
 - b. Not detected – perform the next step.
3. Check the power cable. For machines with a:
 - a. Generic power cable (P/N 150411-04) – Disconnect the power cable at the 6 pin Deutsch connector. Measure the voltages between pin 1 (Batt+) and pin 2 (GND) for a constant battery voltage, 12V or 24V. Also measure the voltage between pin 3 (Key switch) and pin 2, which should have the same voltage when the ignition switch turns on. If the expected machine voltage is not detected, check the fuse on the appropriate supply wire. Replace as necessary with the correct fuse (Key switch is ATC 5A Blade type, Batt+ is ATC 15A Blade type). Also check the ground wire connection to the machine chassis – clean or repair if needed. Replace power cable if needed.
 - b. Machine-specific power cable – Refer to the wiring schematic for your machine. Trace the power wiring back from the TD5x0 display and machine power cable. Measure voltages at each mating connector pair to identify possible cable faults. Replace the cables and harnesses as required.

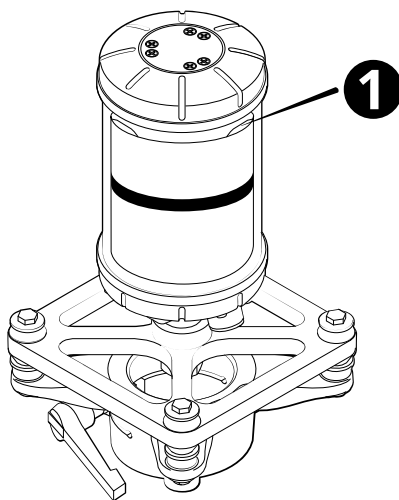
7.2.3 Other electrical issues

Possible issue	Possible issue and resolution
The app is slow on my BYOD	Too many devices may be connected to the Wi-Fi channel, consider using: • a different Wi-Fi channel on the EC520 • a display with a cable connected
The system status tile on the dashboard is red and the Start button is not available.	The GNSS cable is disconnected. Confirm that the GNSS cable is connected.
The GNSS power LED light is off.	No power to the GNSS. See 7.1 Check physical for resolutions.
The operator sees the GNSS ERROR No GNSS Position message.	No power or communication to the GNSS receiver. Check the power and communication connections. No power to the GNSS. See 7.1 Check physical for resolutions.
The LED status indicators on the LR410 laser receiver are not illuminated	No power to the LR410. No power to the GNSS. See 7.1 Check physical for resolutions.
NOTE –The LR410 is unavailable on some machine types.	

7.2.4 MT9xx machine target status indicators troubleshooting

The MT9xx is unavailable on some machine types.

The MT9xx machine target has four status indicators, located at the top of the target (❶). The status indicators show if the unit is receiving power and functioning correctly.



The indicator patterns showing the machine target status are as follows:

Indicator flashing pattern	Meaning
Slow flash (0.1 second on, 0.9 seconds off)	Normal operation
Not illuminated	No power
Fast flash (0.1 second on, 0.1 second off)	Power up (approximately 0.5 seconds), otherwise low battery (<9 VDC)
Blink (3 seconds on, 0.1 second off)	Hardware fault

7.2.5 GS5x0 sensor troubleshooting

If the system reports a GS5x0 sensor error, perform the following troubleshooting steps:

1. On the Web Interface's Onboard Devices screen, select the Recheck button to re-check the error. If it clears, the issue was temporary.
2. Review the error message on the System Status screen:

Error message	Possible issue and resolution
Not Found	The system cannot detect the sensor at all. Possible causes are: — a cable error — incorrect CAN termination — sensor failure If the sensor is not found directly after installation, check that the harness has the correct CAN terminators. If the sensor is not found during operation, check the cables connecting the sensor.
Sensor issue detected	The sensor is communicating, but there is an error. With system power on, use a multimeter to check that the voltage to the sensor is correct: — The voltage between ground and the CAN lines should be 2 to 3 V. — 0 V indicates a short to ground. — Machine V (for example, 24 V) indicates a short to power. In either case, look for a crushed or cut wire. If the error occurred during machine start-up, the cause may be a temporary voltage drop while the engine draws power.

7.3 Check configuration

Symptom	Possible issue and resolution
User preferences are lost after transferring files from a USB flash drive	When transferring files from a USB flash drive to the machine, all files on machine that have the same name as files on the USB flash drive will be overwritten with the USB flash drive files. This includes the user preferences file <code>userdata.pref.xml</code>, which saves the settings; for example, the text items that are configured. To keep the user preferences saved on a machine, either: - Perform a File Transfer of the EarthworksData to the USB flash drive first, then perform a File Transfer from the USB flash drive to the machine OR - Untick the EarthworksData box when transferring files from the USB flash drive to the machine NOTE –The <code>userdata.pref.xml</code> file is stored at ProjectLibrary > EarthworksData > [Machine Name Folder]
Machine parts are not seen in the Operator Interface and the Web Interface	Parts may have power supply issues. See Chapter 7.2, Check electrical
The license tile on the dashboard is red	A license is missing: Check that the required licenses for the current machine configuration have been purchased and are installed on the EC520, TD5x0 and/or GNSS receivers.
Licenses installed on the display do not appear in the Licenses screen in the Web Interface	Open the Operator Interface and allow it to upgrade itself to the same version as the server.
License will not install	Wrong license for the device. Compare the serial number on the license file name against the device serial number, they must match. The license file is corrupt. Go to the Trimble store and download the file again and install again.

Symptom	Possible issue and resolution
The required licenses have been purchased and installed but the above troubleshooting steps have not resolved the issue.	Isolate and restart the machine, the license should appear. Contact support.
All .sg6 files are invalid or corrupt after upgrading licenses	The licenses on the TD5x0 do not automatically upgrade. Open the Operator Interface to start the upgrade.
My Operator App reports a missing device.	The EC520 and device did not communicate correctly, or the installer did not configure the device correctly. To resolve: 1. Open the onboard software. 2. Open Dashboard > System Status and tap Recheck. 3. Check the status indicators beside each device. If the indicator is a. Green - resolved b. Red - power cycle the machine and Operator App If power cycling does not resolve then check the device and harness for damage. If no damage, see 7.2 Check electrical.
My device is not listed on the Onboard Devices list	The missing device is disconnected. Connect the missing device.
A low accuracy error message appears on the display and there is no cut/fill guidance. The left GNSS data link LED is off.	No correction messages are being received from the base station. Check that the correct correction source is selected in the Operator App.
Depth and slope mode only available on my excavator. I cannot select design mode.	No 3D Design License. Contact your dealer. Confirm GNSS is on in the System Status tile. Turn on if off. Power cycle the machine if you have licenses on the display.

Symptom	Possible issue and resolution
Single GNSS on Cat Next Gen excavator is not working correctly.	Single GNSS on Cat Next Gen excavators requires information from the machine system to function. Faults on the machine side may prevent Single GNSS on Cat Next Gen excavators from functioning. If a fault occurs, it will show on the machine display, not the TD5x0. In particular, you must address faults from the machine swing sensor (4326-XX) and machine travel controls (5332-XX or 5352-XX) for correct operation.
In the Web Interface, I can start Configure > Install Assistant > Mast Measure-up but I cannot complete the center of rotation (COR) measurement. The calibration process stalls and I can't complete the Mast Measure-up section.	The machine is not on stable, flat ground. Or the GNSS errors are too large or do not have an RTK position. Try one of the following: • Open Home/Machine Summary, to confirm RTK accuracy of 10 mm horizontal, 20 mm vertical • Place the linkage as close to the ground as possible before starting to spin. • Shift the machine to flat, stable ground and restart the calibration process.
In the Web Interface, when measuring the attachment position, I cannot start the wizard because the GNSS errors are too large or do not have an RTK position.	The GNSS receiver does not receive corrections. Try one of the following: • Ensure you used the correct GNSS measure-up cable (P/N 150775-46). Using incorrect cables or extension cables may increase resistance of the CAN bus. • Note the resistance on the CAN bus and compare to working systems. • Ensure the measure-up cable is attached in the correct location as specified in the instructions on the screen.

Symptom	Possible issue and resolution
Cannot synchronize with Connected Community	Check the signal strength icon in the status bar. The system may have no connection, a poor or intermittent connection, poor cell reception, or poor Wi-Fi. Check the Web Interface for Connected Community settings and any error messages, and check that Connected Community is enabled: Network > Cloud Services Check the Web Interface for gateway device messages: Network > Gateway Confirm the data APN setting is set on the SIM card. Close the Operator App and load a web page on the Android screen.
In the Web Interface, I can open Configure > Install Assistant > Mast Measure-up, but I cannot start the wizard because the GNSS errors are too large or do not have an RTK position	Verify that the machine's GNSS positioning is loaded and working. The system may not be receiving corrections from the base station. To resolve: 1. Open the Web Interface, exit the Install Assistant > Mast Measure-up screen, open Network > GNSS Correction Source, and check that the correct correction source is selected or add a Correction Source. 2. Complete Configure > Install Assistant > Mast Measure-up.
My user name is not on the list on the log in screen.	To resolve:
In the TD5x0, I touch the icon to open the application. The log in screen appears, but I get the error message, "No operator configured for this machine"	1. In the Web Interface, open Advanced > User Permission, and add the names of users who need permission to open the onboard software. 2. Select Operator for the needed user and save. TIP – Installers - create operator and administrator permissions immediately after completing the Configure section for each machine.
The precisions on the Web Interface change from green to red to green during a 3D mast measure-up using GNSS receivers.	No issue. Wait for the measurements to finish recording, then continue with the measure-up.

Symptom	Possible issue and resolution
The operator sees the "Radio does not support UTS" message	The on-machine data radio does not support the data transfer protocol used by the UTS instrument. Set the on-machine data radio to the 2400 MHz radio band.

To check:

- the attachment measure-up process, open Configure > Attachments.
- the measure-up process, open Configure > Install Assistant > Measure-up.
- the 3D sensors measure-up process, first check the accuracy of the body sensor calibration process by checking the machine pitch and roll text items in the software of the machine against a calibrated digital level. Next, confirm you haven't changed the GNSS receiver mounting locations. Finally, redo Configure > Install Assistant > Mast Measure-up.

LED	Cause	Resolution
The green LED is off.	Suspect a power supply problem.	Check the power supply to the valve module. Use the connection diagram and schematics to check that the valve module is connected correctly.
The green LED is flashing.	Suspect incorrect termination.	Check all terminators are present and connected. The resistance between CAN hi and CAN lo must be about 60 Ω . Otherwise, suspect CAN driver issues.
The green LED is on continuously.	Suspect incorrect CAN bus connection.	Check all connections. Check that you have used the correct cables.
The red LED is either flashing or on continuously.	Suspect a valve module fault.	Replace the valve module.

7.3.1 Check compatibility of GNSS receivers

GNSS receivers within the same series may be paired. You cannot pair receivers from two different series (for example, an MS97x receiver cannot be paired with an MS99x receiver).

The compatibility of receiver pairs depends on the machine type and the firmware that is installed on each receiver.

Blade-mounted receivers - permitted combinations

Receiver	Compatible receiver combination	Firmware requirement
MS992	MS992	firmware must match
MS995	MS995	firmware must match
MS996	MS996	firmware must match

NOTE –The receiver firmware must match or be greater than the minimum required version.

Cab-mounted receivers - permitted combinations

	MS952*	MS955*	MS956*	MS972	MS975	MS976	MS992	MS995	MS996
MS952*	firmware must match	firmware can be different	firmware can be different	N/A	N/A	N/A	N/A	N/A	N/A
MS955*	firmware can be different	firmware must match	firmware can be different	N/A	N/A	N/A	N/A	N/A	N/A
MS956*	firmware can be different	firmware can be different	firmware must match	N/A	N/A	N/A	N/A	N/A	N/A
MS972	N/A	N/A	N/A	firmware must match	firmware can be different	firmware can be different	N/A	N/A	N/A
MS975	N/A	N/A	N/A	firmware can be different	firmware must match	firmware can be different	N/A	N/A	N/A
MS976	N/A	N/A	N/A	firmware can be different	firmware can be different	firmware must match	N/A	N/A	N/A
MS992	N/A	N/A	N/A	N/A	N/A	N/A	firmware must match	firmware can be different	firmware can be different
MS995	N/A	N/A	N/A	N/A	N/A	N/A	firmware can be different	firmware must match	firmware can be different
MS996	N/A	N/A	N/A	N/A	N/A	N/A	firmware can be different	firmware can be different	firmware must match

*MS95x models are compatible with and require the use of Zephyr™ 3 Rugged antennas.

NOTE –The receiver firmware must match or be greater than the minimum required version.

NOTE –N/A = incompatible combination.

NOTE –Two receivers with the same model number connected to the system must have the same firmware version. Two compatible receivers with different model numbers can have different firmware versions.

7.3.2 Replace GNSS receivers

Permitted combinations of MS97x and MS99x receivers, as listed above, are interchangeable when mounted on quick release brackets. You do not have to redo the measure-up if you replace one pair of these receivers with another.

NOTE –A machine on which the measure-up was performed with MS95x receivers installed does not support running with MS97x or MS99x series receivers. You will have to redo the measure-up if you replace the MS95x receivers with MS97x or MS99x receivers.

7.3.3 License troubleshooting

Symptom	Possible issue and resolution
The license tile on the dashboard is red	A license is missing: Check that the required licenses for the current machine configuration have been purchased and are installed on the EC520, TD5x0 and/or GNSS receivers.
Licenses installed on the display do not appear in the Licenses screen in the Web Interface	Open the Operator Interface and allow it to upgrade itself to the same version as the server.
License will not install	Wrong license for the device. Compare the serial number on the license file name against the device serial number, they must match.
	The license file is corrupt. Go to the Trimble store and download the file again and install again.
The required licenses have been purchased and installed but the above troubleshooting steps have not resolved the issue.	Isolate and restart the machine, the license should appear.
	Contact support.
All .sg6 files are invalid or corrupt after upgrading licenses	The licenses on the TD5x0 do not automatically upgrade. Open the Operator Interface to start the upgrade.

7.4 Check accuracy

Symptom	Possible issue and resolution
The display shows GNSS Error, Autonomous GNSS	Corrections from the base station are not being received. Check the Correction Source selection is correct, if the problem persists check that the base station is still broadcasting corrections. If you are using an internet delivered correction source, check that you have an internet connection and either cell or Wi-Fi in the area you are working in.
While operating the machine, an orange alert banner appears with the message GNSS precision outside the set tolerance. I looked at the text bar and the cut/fill values are not there. The left GNSS data link LED is off.	The data coming from the GNSS receiver about GNSS Precision Tolerance is outside the range set in the software. This could be due to loss of corrections from the base station or due to obstructions, such as trees, buildings, etc. Possible resolutions: 1. In the Operator App, open System Settings > GNSS Precision Tolerance, and check the H and V precision values and confirm the appropriate mode is selected for the task. 2. If you are using a radio correction source, consider using a repeater to improve radio signal strength. 3. If you are using an internet-delivered correction source (VRS, IBSS, NTRIP) check the signal strength icon in the status bar. The system may have no connection, a poor or intermittent connection, poor cell reception, or poor Wi-Fi. 4. Ensure the GNSS receivers have a clear view to the sky. 5. The radio or base station may be turned off. Confirm that the base station is outputting corrections by confirming that other GNSS devices on site are receiving corrections. 6. Confirm that the data link LED is flashing and move the machine: ─ Away from trees, buildings, or other tall obstacles. ─ Closer to the site base station.

Symptom	Possible issue and resolution
After I finish configuring my machine, in the Web Interface there are no values in Monitor > Machine Positions.	You have not loaded a Project so the software is missing the required coordinate system for populating these fields. To resolve: 1. In the Operator App, create or select a project in Dashboard > Job Setup. Loading a project loads a coordinate system.
The operator needs to confirm connection strength for TLL/VSS/NTrip	1. Open the Operator App and go to System Settings > System Status 2. Press and hold GNSS Receiver - Left 3. Wait for the status screen to open 4. Confirm: a. Integrity $\geq 90\%$ b. Latency ≤ 2 sec
My design does not appear on the TD5x0 after transferring files via a USB flash drive	Confirm that the USB flash drive is fully plugged in with the light flashing. Some USB flash drives have casings that are too wide for the TD5x0 USB flash drive port. Confirm that the file is on the USB flash drive by opening the USB flash drive on a computer. NOTE –To ensure the file structure on the USB flash drive is optimally set up for the system, insert a blank USB flash drive into the TD5x0, open Trimble> System Settings > File Transfer.
The operator sees the GNSS ERROR No GNSS Position message.	No valid site calibration or ACS (Automatic Coordinate System) loaded.
Machine is now inaccurate after restarting the machine but was accurate before machine shut down	The GS520 grade sensors must be at operating temperature. In colder temperatures, warm up the Earthworks system for at least 10 minutes after turning on power. The GNSS is connected to the wrong base station or the base station may have been moved.

Symptom	Possible issue and resolution
Machine now has Pass Matching issues while in Autos but was accurate previously.	The system has an error. When the machine is working in Autos, determine if: • The issue is constant, determine if the constant issue occurs when working on: ─ Level ground and slopes - redo the Body and Measure-up sections ─ Slopes only - redo the Pass Match Adjustment section • The issue is not constant, especially when the blade drifts off grade while making a pass and drifts back to grade when the machine is stopped - redo the Sensor Correction Check section. The Sensor Correction Check requires machine access to a level surface and a slope of at least 10%.
The initial sensor initialization is unsuccessful	The sensors have a bias. Perform a Sensor Correction Check.
Occasionally and in some applications, cut/fill drifting occurs on machines with body sensors only	• If you perform the check within 100 operating hours of the measure-up occurring, the system will update the measure-up using the corrected sensor values. • If you perform the check after more than 100 operating hours, perform a new measure-up.
Occasionally and in some applications, cut/fill drifting occurs on machines with blade mounted sensors	

To check:

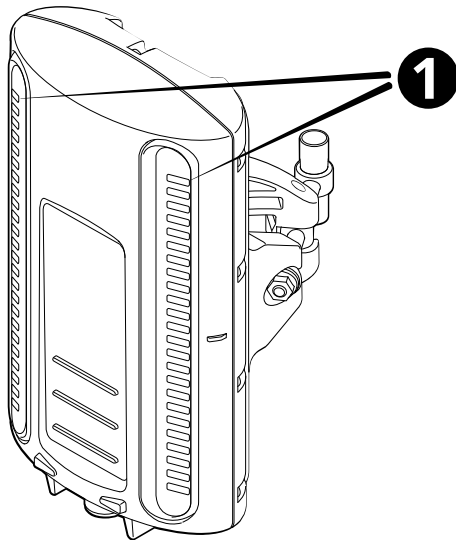
- the attachment measure-up process, open Configure > Attachments.
- the measure-up process, open Configure > Install Assistant > Measure-up.
- the 3D sensors measure-up process, first check the accuracy of the body sensor calibration process by checking the machine pitch and roll text items in the software of the machine against a calibrated digital level. Next, confirm you haven't changed the GNSS receiver mounting locations. Finally, redo Configure > Install Assistant > Mast Measure-up.

7.4.1 2D-Sensor troubleshooting

The ST400 and LR410 are unavailable on some machine types.

Check LR410 status indicators

The LR410 laser receiver has two sets of LED status indicators (❶).



LED pattern	Meaning
Not illuminated	No power
Slow flash (0.4 seconds on, 1 second off)	No strike
Flash (0.2 seconds on, 0.2 seconds off)	Strike detected above center of the receiver
Fast Flash (0.1 second on, 0.1 second off)	Strike detected below center of the receiver
On continuously	Strike detected at the center of the receiver

7.4.2 2D system troubleshooting

Symptom	Possible issue and resolution
Guidance lost due to laser interference	An LED or similar light source on the body of the machine is interfering with the operation of the laser receiver. Raise the laser transmitter and receivers, if possible, until guidance is restored.
When benching a laser receiver, the operator gets the error message "No laser strike detected"	Too much variation in the output of the machine's angle sensors. Ensure the machine is stationary and then bench it.

Symptom	Possible issue and resolution
When benching a laser receiver, the operator gets the error message "Not enough laser strikes"	Check transmitter line of sight and stability. On sites with multiple laser transmitters, check that the receiver is not detecting a second laser strike.
When benching a laser receiver, the operator gets the error message "Too much blade movement"	Too much variation in the output of the machine's angle sensors. Ensure the machine is stationary and then bench it.

7.5 Check performance

To check the machine performance, refer to the tests in the *validation guide*.

Refer to the Autos Performance troubleshooting decision tree available on [Partners](#) for other performance checks to consider.

7.6 System errors

This section describes some system errors that may occur.

The Operator App shows simplified error and warning messages.

For detailed error and warning messages, see the Web Interface's Monitor > Program Log screen.

7.6.1 GNSS receiver errors

This section describes possible errors reported by a GNSS receiver.

To view any active GNSS errors, select the Monitor > GNSS Details screen in the Web Interface. Active errors appear at the top of the screen.

Error Code	Meaning	Action
1	Not enough satellites to do any kind of fix.	Check for obstructions between the antenna and the satellites.
2	PDOP too high to do autonomous fix.	Check for obstructions between the antenna and the satellites.
3–63	Conditions too poor for any positioning.	Check for obstructions between the antenna and the satellites.
64	Wait a few seconds.	The issue may resolve.
65	No base station data available.	Check the radio configuration.
66	Base station coordinates unknown.	Check the radio configuration.

Error Code	Meaning	Action
67	Not enough satellites to do differential fix.	Check for obstructions between the antenna and the satellites.
68	Valid reference satellites could not be found.	–
69	Base station is not tracking the same kind of satellite code that rover is.	Check the base configuration.
70	PDOP is too high to do differential fix.	Check for obstructions between the antenna and the satellites.
71–72	Internal error or misconfigured RTK engine.	Contact your support channel.
73	Wait a few seconds.	The issue may resolve.
74–75	Too many satellites blocked or geometry too poor for a precise position.	Check for obstructions between the antenna and the satellites.
76	Internal error or misconfigured RTK engine.	Contact your support channel.
77	Exceeded baseline limit set in option blocks.	–
78	No initial RTK rover position.	The issue may resolve.
79	Base station tracking is weak.	–
80	Tracking is too weak or there is too much multipath for a precise position.	Check for obstructions between the antenna and the satellites.
81	Missing multi-band reference information.	Check the base configuration.
82	Missing multi-band rover information.	The issue may resolve.
83	Contact your support channel to review receiver options.	Contact your support channel to review receiver options.
84–127	Conditions too poor for differential positions.	Check for obstructions between the antenna and the satellites.
128	Could not find sufficient L2 phase.	Check for obstructions between the antenna and the satellites.

Error Code	Meaning	Action
129	RMS value for RTK fixed position is too high.	Check for obstructions between the antenna and the satellites.
130	Wait a few seconds.	The issue may resolve.
131	Internal error or misconfigured RTK engine.	Contact your support channel.
132	Insufficient satellites to carry out search.	Check for obstructions between the antenna and the satellites.
133	Internal error or misconfigured RTK engine.	Contact your support channel.
134–135	Too many satellites blocked or geometry too poor for a precise position.	Check for obstructions between the antenna and the satellites.
136	Propagation time exceeded limit.	Check for obstructions between the antenna and the satellites.
137	DOP of search satellites was too high.	Check for obstructions between the antenna and the satellites.
138	Internal error or misconfigured RTK engine.	Contact your support channel.
139	Reference station is too far away to – initialize.	
140–142	Too many satellites blocked or geometry too poor for a precise position.	Check for obstructions between the antenna and the satellites.
143	Internal error or misconfigured RTK engine.	Contact your support channel.
144	Contact your support channel to review receiver options.	–
145–150	Tracking is too weak or too much multipath for a precise position.	Check for obstructions between the antenna and the satellites.
151–254	Conditions too poor for precise positions.	Check for obstructions between the antenna and the satellites.

7.6.2 WorksOS errors

This section describes possible errors encountered when using the WorksOS online service.

Error Code	Meaning	Action
3047 3100	Ground surface download is not configured correctly: unable to identify device.	Check that your device is registered.
3102 3103	Ground surface download is not configured correctly: device account issue.	The system cannot locate the WorksOS account for the device. There is more than one active WorksOS account for the device.
3048 3105	Ground surface download is not configured correctly: no office project found for this device.	Check with your office data manager that the device is assigned to the relevant WorksOS project.
-5 3019 3020 3037 3049 Others	Ground surface download error: contact support.	There was an unexpected issue. Contact support for help.
3017 3124 -3	Unable to connect to server. The system will retry shortly. Contact support if this does not resolve.	The system is unable to connect to the server. The system will retry. Check the internet connection. If the system continues to report an error, contact support.
-	Cannot contact server. Ensure your system is connected to the internet.	Check the internet connection.
-	Download failed: system will retry shortly.	The download failed or was interrupted. The system will retry after a brief pause.
-	There is no office project at this location. Contact your office data manager.	You are outside of any WorksOS project boundary. Check with your office data manager that the WorksOS project for your location is available.
3021	System error. Contact support.	The system is using UTS navigation but there is no .cal file in the project. Add a .cal file so the system can identify the correct project in WorksOS.

7.6.3 License errors

The Status text at the top of the Licenses screen reports any license errors. Possible errors are:

Error Code	Meaning	Action
Network error 6	There is a DNS error.	Check your internet connection. If the issue persists, contact support.
Network error 28 / -1	There is a timeout communicating with the server.	
Other network errors	There is a network error.	
Install error 5	An install is already in progress.	Wait 30 seconds and retry. If the issue persists, restart the system.
Install error 6	The device is not found.	Check the device connection and license service state. If the issue persists, contact support.
Install error 7 / 11 / 16 / 17	The display's time is incorrect.	Correct the display's time.
Install error 10	The device's operating system does not support cloud licenses.	Update the device's operating system to a version that supports cloud licensing.
Install error 14	The uploaded license is an option key, but the device requires a cloud license.	Obtain a cloud license.
Other install errors	The installation failed.	Retry the installation. If the issue persists, contact support.
HTTP error	There was an unexpected issue with proxy or network settings.	Retry later. If the issue persists, contact support.
All other errors	An error has occurred.	

7.6.4 Device errors

To view the status of hardware devices in the system, see the Web Interface's Monitor > Onboard Devices screen.

Each device connected to the system has an icon that shows the current status of the device:



The potential states are:

Symbol	Detail
	The device is functioning with data flowing correctly.
	The device is connected but may be in one of the following states: • Unreliable data • Configuring • Configuration failed • Firmware update required • Not required in the current machine configuration
	• Faulty: The sensor may have a non-recoverable fault. • Not Found: The system cannot detect the device. • Issue: There is a problem with the device. You can probably resolve it without replacement.

For more information, click the expand arrow of a device.

GS5x0 status descriptions

Field	Detail
Device Connection	Shows whether the sensor is connected over CAN. One of:
	 The sensor is correctly connected.
	 None: The sensor is required but not connected over CAN.

Field	Detail
Device Status	Shows whether the sensor hardware is working correctly. One of:
	 The sensor is working correctly.
	 None: The sensor is required but not connected over CAN. Low Voltage: The sensor is reporting low voltage. Replace Sensor: The sensor is reporting a non-recoverable error.
Firmware	Shows whether the sensor's firmware meets the minimum requirement:
	 The firmware meets the minimum required version.
	 Update Required: Firmware does not meet the minimum required version.

7.6.5 Design validation

The system supports designs in LandXML, .vcl or .dsz format. When a design is added to the system, the system validates the file to ensure it is valid. You cannot use the file until validation is complete.

When a design is queued for validation, the system shows the validating icon .

When the system is validating the design:

- The system displays the validating icon  and a time on the Job Setup screen beside the design's name in the Design File menu.
- The time is an estimate of how long the validation will take.
- A warning message displays:

On-screen Message	Description
 File > Validation in progress...	Wait for the system to complete the validation process.

NOTE –If you load a design, the system will pause validation of other designs.

If a design was received automatically from WorksManager, the error or warning is also recorded in the WorksManager system.

Validation timing

The time that the system takes to validate your designs varies depending on the complexity of the design:

- In the Design File drop-down, the system shows a countdown of the estimated time remaining for the design that is being validated.
- To validate a specific design ahead of other designs, select it in the Job Setup screen.

NOTE –If you introduce a project that contains many complex designs, the time to validate them may be significant.

Troubleshooting designs

Files within a project can fail validation for a number of reasons. This section describes some causes. Validation occurs for several file types within a project (not just designs).

TIP – To correct a design that fails validation, you must adjust the design in the office software that generated it (such as Trimble Business Center). Therefore, the information in this section may be most relevant to site engineers or data managers.

TIP – Highly complex designs can fail validation because they require too many system resources. Smaller designs (for example, less than 20 MB) are less likely to cause problems.

Validation troubleshooting

You can receive 2 types of validation message:

- A red cross : the design cannot be used. The file fault will need correcting in the design tool.
- An orange exclamation mark : the design can be loaded, but there is a warning. It may not display fully or provide guidance as expected.

The Operator App provides basic guidance on why a design fails validation. This section describes how to get more detailed information.

When the system validates a design, it creates a text file called *ProjectDiagnostics.txt* in each project's folder that records details of the validation. The contents of this file can help you to locate the cause of the failure and correct it.

NOTE –The file is only generated for new validations, so it won't exist for designs validated on earlier versions of the software.

To obtain the file:

1. Use the File Transfer screen to export the project to a USB drive.
2. On a PC, navigate to the *ProjectDiagnostics.txt* file in the project.
3. View the file in Notepad and scroll to the section for the design that failed validation.
4. Look for the following strings:

- Validation State
- Validation Warnings

Here is an example from a design that failed validation. *ProjectDiagnostics.txt* states:

- Validation State: Invalid
- Validation.Warnings.0: Linework extends outside design boundary (400x400km)

This design was invalid because a line element extends beyond the design boundary. There may be more than one reason why a design fails validation. The validation file records the first 5 reasons for failure.

For designs that are usable but generate an orange warning because an element is missing , you can also find detailed validation information in the Web Interface's Monitor > Program Log screen. Look for warnings relating to <NavigationProcessorComponent>. For example:

WARNING - <NavigationProcessorComponent> Validation warning: Linework extends outside design boundary (400x400km)

NOTE –The design may display with expected elements missing. If the element you require is not present, the file issue will need correcting in the design tool.

Issues that stop the system from loading the design include:

On-screen Message	Description
 Issue > File Missing > Check File Structure	You cannot load the design because the system could not find the file. This issue can also occur if the file is present but is 0 kB.
 Issue > Unrecognized Format > Check Design	You cannot load the design because the system does not recognize the file format.

On-screen Message	Description
 Issue > Invalid File > Check Design	You cannot load the design because the file is invalid. This may be because: • The file does not conform to the design schema. (This condition does not apply to LandXML.) • Units of measure are not defined in the file. • The units that are defined are unsupported. • The maximum number of line elements is exceeded.
 Issue > Invalid File > File Size Too Large > Check Design	You cannot load the design because the file's size is too large: • For .vcl files, the limit is 150 MB. • For LandXML files, the limit is 50 MB. • For .dsz files, there is no size limit.
 Issue > Surface Too Complex > Check Design	There are too many triangles in the design's TIN model (more than 65,000 in one page).
 Missing project coordinate system file	The system cannot find a .cal file for this project. Check the project's \OfficeData\ folder.
 No Measured Data selected	Select a Measured Data file in Job Setup screen.

Issues where there is a problem with the design, but you can still load it, include:

On-screen Message	Description
 Avoidance zones do not work with a 2D positioning source	The selected project contains one or more avoidance zones, but cannot display them because you are using a 2D positioning source.
 Design File > Surface Issue Design loaded, but only valid TIN surfaces will show. Check the design file.	The system can load the design, but there is an issue with one or more surfaces: • The surface may be missing an element. • The surface is not a valid TIN model.

On-screen Message	Description
 Design File > Surface Issue Design loaded, but one or more surfaces contain missing or invalid data and may be unusable. Check the design file.	The system can load the design, but there is an issue with one or more surfaces. Causes include: • Missing faces or points • Invalid points • Invalid face • Face refers to non-existent points • System cannot process the surface TIN model
 Design File > Linework Issue Design loaded, but one or more layers contain missing or invalid linework data and may be unusable. Check the design file.	There are many possible causes of this error. Refer to the <i>ProjectDiagnostics.txt</i> file for more detail.

On-screen Message	Description
Linework elements unsupported in LandXML • Line is not 2D or 3D • Curve is not 2D or 3D • Curve has inconsistent radius • Curve length does not match provided length • Curve has undefined bearing or arc length • Invalid point list • Point list has incorrect number of values • Line color is unknown format • Point has invalid coordinates • Point has no name • Curve has invalid rotation • Curve has invalid or no center coordinate • Start/end dimensions do not match • Line has wrong contents • Alignment does not have a name • Alignment has no staStart or length • Alignment total length does not match	
• Line only has one horizontal element • Error converting string to double • cgPoint contains no value or reference • Invalid spiral radius • Cannot find point in layer • No layer exists • Circular referencing • Invalid point • Cannot find point in lookup table • Point name should be unique • Feature tag has empty label • Feature tag has empty value • Invalid PI dimension • Spiral start and end point is neither 2D nor 3D • Linework contains bad data • Unsupported linework type	

On-screen Message	Description
Linework elements unsupported in VCL:	
• Unsupported linestring elements: – Dependent Location Three Point Arc – Dependent Three Point Arc – Dependent Segment Three Point Arc – Dependent PI Arc with elevation – Offset with useSlope – Offset endpoint with negative along – Endpoint with StationOffset • Unsupported linestring vertical elements: – eStationSlopeToParabola – eStationSlopeFromPI – eStationSlopeFromParabola – eStationSlopeToFromPI – eStationSlopeToFromParabola – eStationSlopeToArc – eStationSlopeFromArc – eStationSlopeToFromArc • Singular best fit arc linestring element • Best fit arc linestring element with the same start and end points • Smoothed polylines • Smooth Curves	• RoadIntersectionLinestring • UtilityLine • Hatch • Moss6D • Linestring with no elements • 3D spirals • Smart text • Decreasing stationing • Point Coordinate Component • Point Coordinate Type • Vertical alignment segment of type "<type>" • Offsets to alignment spirals • Any entities defined on a cutting plane • Any entities defined using a User Coordinate System • Composite surfaces • Only single point coordinates supported, <number> points found • Only alignments with curve type = "arc" are supported • Only alignments with spiral type = "Clothoid" or "Cubic" or "Cubic Parabola" or "NSW Cubic Parabola" or "Half Sine" are supported

Design complexity limits

A design can also fail validation if it contains too many elements. Exceeding the following limits can cause a design to fail validation:

Element	Limit
Too many lines or linework elements (including curves)	65,534 lines in one page.

Element	Limit
Dense surface	Avoid using surfaces where the TIN triangles are too small, e.g. under 0.01 m ² per triangle or has more than 65,000 triangles in an area of 50 m x 50 m.
Too many points	160,000 points.
Oversized TIN triangles	TIN triangles with 40 km sides. If you require extremely large TIN triangles, use .dsz format.
Avoidance zones	800 segments (or sides) per polygon. The system supports multiple avoidance zone polygons. For best performance: • Try to keep individual polygons below 500 segments. • Break up a large or thin polygon into multiple polygons with similar dimensions.

In addition, use of the following may cause issues within a design:

- Contains a spiral, horizontal cubic, or NSW spiral
- 3D lines of curves
- Complex vertical segment arcs with horizontal spirals
- Ferguson linestring elements
- Excessive text/labels
- Excessive line names
- Excessive layers
- Design triggers many warnings/information notes

7.6.6 TCC (Connected Community) errors

This section describes possible errors encountered when using the TCC online service.

Error Code	Meaning	Action
0	Success: The files synced successfully with TCC.	N/A
2	Canceled: Either the technician canceled the TCC file sync, or an event occurred that canceled the sync (for example, an operator logging into the Operator App).	If you did not cancel the sync, consider whether an event (such as operator login) caused the cancellation. Repeat the sync.

Authentication errors:

Error Code	Meaning	Action
10	Failed login: the login attempt failed because of an account issue	Ensure that you are using the correct login details and that your subscriptions are valid.
11	No credentials	TCC sync failed because the machine file is missing a required machine name or organization name.
12	Machine name not set	

Networking errors:

Error Code	Meaning	Action
20	Server connection failed	Check the internet connection is active and the TCC configuration details are correct.
21	HTTP host not resolved	
22	Network unreachable	Check if the network is blocking data.

Error Code	Meaning	Action
23	HTTP request failed	When the grade control system attempted to communicate with TCC, the online service returned one of the standard HTTP error codes (you can review the meaning of your specific code online). Common codes include: • 400 – Bad request. • 401 – Unauthorized. • 403 – Forbidden. • 404 – Not Found: The requested URL does not exist. • 429 – Too many requests. • 502 – Bad Gateway: When you connect to a website, your request is routed through a string of servers. A server at some point between your system and TCC has rejected the request. The error may resolve at a later point simply because a different set of servers is used to reach TCC. It may also be caused by your system configuration. • 503 – Service unavailable. • 504 – Gateway timeout. • 511 – Network authentication required.

Server errors:

Error Code	Meaning	Action
30	Remote files do not exist	The file that you are attempting to download from TCC does not exist (for example, because it was deleted, making your list of files out of date).
31	Should not continue	The server reported an error.
32	List remote directory	

Error Code	Meaning	Action
33	List file returned multiple files	An error occurred because the system requested details of one file but received details for multiple.
34	Create remote directory	The server reported an error.
35	Cannot uplink zero-sized file	An error occurred because the grade control system tried to upload a zero-sized file to TCC.
36	Rename remote	The server reported an error.
37	Delete remote	
38	Delete remote directory	
39	Delete remote lockfile	
40	Get remote modify time	
41	Parsing JSON	The server sent a response that is invalid.
42	Uplinking chunk to remote	
43	Invalid file space ID from remote	

Local file system errors:

Error Code	Meaning	Action
60	Insufficient space	Free up space on the EC520.

Error Code	Meaning	Action
61	The grade control system failed to rename a file locally.	The local system reported an error.
62	The grade control system failed to delete a local file.	
63	The grade control system failed to create a directory locally.	
64	The grade control system failed to delete a local lock file.	
65	The grade control system failed to open a local file.	
66		
67	The grade control system failed to update the time on a downloaded file.	

Other errors:

Error Code	Meaning	Action
80	The system is already trying to log in.	Wait for a short period.
81	The system is already trying to sync with TCC.	Wait for a short period.
83	Not permitted.	–

TD5x0 Android Recovery Mode

In this chapter:

- ▶ Recovery Mode
- ▶ Reset a TD510 or TD520 Android password
- ▶ Reset a TD540 Android password
- ▶ Exit the No Command screen
- ▶ Exit Recovery Mode menu

8.1 Recovery Mode

Recovery Mode provides an option to clear a TD5x0 hard drive. This option is useful for resetting a TD5x0 Android password. This option resets the Android device password only and does NOT reset the Trimble® Earthworks password.

8.2 Reset a TD510 or TD520 Android password

To reset the Android password on a TD510 or TD520:

1. Turn on the display. Press and hold the power button to open an options menu.
2. Tap Recovery and then OK on the Recovery Mode box.
3. Wait for the No Command, or dead Android, image to display.
4. Press simultaneously the power and volume up button a few times until the Android Recovery menu opens.
5. Follow the on-screen instructions to select wipe data/factory reset.
6. Wait for the display to reboot and enter a new password.

NOTE –Wipe data/factory reset erases all files, such as design files, from the display's hard drive. All installed apps remain.

8.3 Reset a TD540 Android password

To reset the Android password on a TD540:

1. Turn off the display. Press and hold the power and volume down buttons at the same time until the display turns on.
2. Wait while the device image displays.
3. Press the volume up button for five seconds and release. This makes the menu open.

4. Follow the on-screen instructions to select wipe data/factory reset.

NOTE –Tapping the screen also scrolls through the menu items.

5. Wait for the display to reboot and enter a new password.

NOTE –Wipe data/factory reset erases all files, such as design files, from the display's hard drive. All installed apps remain.

8.4 Exit the No Command screen

To exit the No Command, or dead Android, screen (before the Recovery Mode menu screen), press the display's power button or leave the display on for two minutes and it will automatically reboot.

8.5 Exit Recovery Mode menu

To exit the Recovery Mode menu, follow the on-screen instructions to select reboot system now.

Supported third party tablet installation

In this chapter:

- ▶ [Overview of third-party tablet installation](#)
- ▶ [Third-party tablet installation](#)

9.1 Overview of third-party tablet installation

9.1.1 Purpose of this guide

The purpose of this guide is to enable the use of supported third-party tablets with the Earthworks system.

9.1.2 Supported third-party tablets

To run this release of the product on a third-party tablet, use a tablet that meets the following requirements:

- OS: Android 6 to 14
- Supported graphics: OpenGL ES 3.0 or higher
- Display: 8 inches or larger, 16x10 aspect ratio
- Google Chrome
- 2 GB RAM

The recommended tablet is the Samsung Galaxy Tab S7 (SM-T870, SM-T875, or SM-T876B).

9.1.3 Prerequisites

Before starting a third-party tablet installation:

- Make sure that the EC520 controller has the required OS and firmware installed, and that the controller's Wi-Fi access point is enabled. For more information, refer to *Earthworks Configure and Connect Guide*.
- It is strongly recommended that you review this entire guide. This will ensure you understand the full installation workflow.

9.1.4 Equipment list

NOTE –For users who supply their own display, the kit still includes a TD5x0 display harness, which must be used for connecting to the AA510:

- A PC
- Optionally, a cable to connect the tablet to a cigarette lighter for power

9.2 Third-party tablet installation

9.2.1 Download the app to the tablet

1. Turn on the tablet.
2. For Android 6 or 7 devices, navigate to Settings > Security > Unknown Source, and allow installation of apps from unknown sources.
3. Navigate to Settings > WLAN and connect to the EC520 controller:
 - The name of the access point is the SSID set during the configuration of the controller.
 - The password of the access point is the password set during the configuration of the controller.

NOTE – Some implementations of the Android operating system do not automatically reconnect to Wi-Fi access points that are not connected to the Internet. If a Network Error message displays after you turn the tablet off, and then turn it on again, check that Wi-Fi is enabled on the tablet, and that the controller is connected.

4. In a web browser, browse to: 192.168.168.1. The Web Interface log-in screen displays.

TIP – Earthworks supports the Google Chrome browser to run the Web Interface.

5. Log in to the Web Interface. The Web Interface Home screen displays.
6. Navigate to Advanced > Install Operator App. The Install Operator App screen displays.
7. Select Install Direct. The Earthworks firmware download begins.
8. On your device, locate the downloaded *launcher.apk* file, and select it.
9. Follow the instructions on the screen. For Android 8 and above, if the install is blocked by Android, set Allow from this source to ON.
10. If prompted, allow the plugin to access photos, media, and files on your device.

9.2.2 Locate and attach the tablet

In the cab:

1. Decide on how to attach the supported third-party tablet in the cab. We do not suggest locations or methods of attachment.

2. Optionally, attach the cable to connect the display to a cigarette lighter for power.

9.2.3 Validate the install

NOTE – To view the log-in screen, make sure at least one operator is configured, and that maintenance lockout is not active. For more information, refer to the *Configuration Guide*.

1. Turn on the tablet.
2. Locate and tap the app icon. If one of the following displays, then the tablet is connected to the EC520 controller:
 - the log-in screen
 - the *No operator configured for this machine* error message
 - the *Maintenance Lockout Active* error message

It is recommended that you install Android updates when available to keep the system up to date.

Tools and other supplies

In this guide:

- ▶ [Dozer measure-up best practices guide](#)
- ▶ [Dozer measure-up and sensor calibration validation guide](#)
- ▶ [Supported third party tablet installation](#)

This guide lists the tools required to complete the commissioning processes.

10.1 Dozer measure-up best practices guide

- Total station, in good adjustment, and with a charged battery
- If required, a data collector with a charged battery, and software
- Two GNSS measure-up target kits (P/N 105568)
- Cleaner/de-greaser
- Cleaning rags
- Marker pen – choose a color easily visible against the color of the blade
- Self-adhesive or magnetic acrylic targets (do not use DR measurement mode, it does not provide the accuracy of measurement required)
- Scissors
- Straight edge
- Digital level

10.2 Dozer measure-up and sensor calibration validation guide

- A total station in good adjustment or a GNSS receiver, with a fully charged battery
- If required, a data collector with a fully charged battery
- A note book and pens

10.3 Supported third party tablet installation

NOTE – For users who supply their own display, the kit still includes a TD5x0 display harness, which must be used for connecting to the AA510.

- A PC
- Optionally, a cable to connect the tablet to a cigarette lighter for power

Torque charts

11.1 Imperial

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

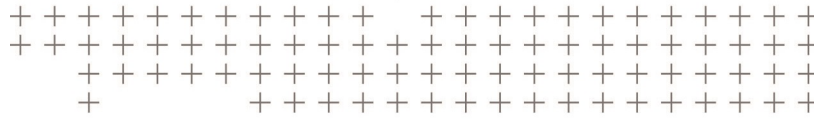
Nominal size (inches)											
	Grade	Torque	1/4	5/16	3/8	7/16	1/2	9/16	5/8	3/4	1"
	SAE 2	FT-LB	6	11	19	30	45	66	93		300
										150	
	SAE 5	FT-LB	9	18	31	50	75	110	150	250	580
		Nm	12	24	42	68	102	149	203	339	786
	SAE 8	FT-LB	13	29	46	75	115	165	225	370	890
		Nm	18	39	62	102	156	224	305	502	1207
	SOCKET HEAD	FT-LB	14	30	50	80	120	175	240	390	960
		Nm	19	41	68	108	163	237	325	529	1302

11.2 Metric

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

Nominal size (mm)										
	Grade	Torque	6	8	10	12	14	16	18	20
	4.6, 4.8	FT-LB	3	8	16	27	43	60	90	120
		Nm	4	11	22	37	58	81	122	163
	8.8	FT-LB	7	17	33	59	100	130	180	280
		Nm	10	23	45	80	136	176	244	380

Nominal size (mm)										
	Grade	Torque	6	8	10	12	14	16	18	20
	10.9	FT-LB	10	25	48	85	130	200	280	400
		Nm	14	34	65	115	176	271	380	542
	12.9	FT-LB	12	29	57	100	150	230	–	–
		Nm	16	39	77	136	203	312	–	–



Notices

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